Cognitive Obesity

Cognitive-Experiential Imbalance as a Unifying Framework

for Depression, Violence, and Information Overload

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Abstract

This paper proposes an integrative framework, grounded in a nutritional metabolism analogy, for the ongoing debate on the relationship between digital media and mental health. The core claim is that the primary predictor of psychopathology is not the absolute quantity of screen time but the balance (additive imbalance) between non-selective cognitive input and experiential processing, with the qualitative classification criterion being sensorimotor loop closure—the presence or absence of bidirectional causal coupling with the environment.

Systematic analysis of 177-country $\times$ 1990–2023 panel data (Study 1) demonstrates: (1) The positive effect of internet penetration observed under country FE alone reverses sign under Two-Way Fixed Effects (TWFE), revealing dependence on common time trends. (2) The strongest predictor of depression prevalence is GDP ($t = 18.27$), likely reflecting a combination of diagnostic capacity expansion (greater clinician density, broader screening) and genuine prevalence changes associated with development-linked stressors, verified through comparison with hard outcomes (suicide rates). (3) Hansen Panel Threshold Regression identifies a threshold of $\gamma^* = 25.7\%$ internet penetration ($F = 213$, bootstrap 95% CI = $[25.7, 32.8]$) for suicide rates: below this level, internet access is protective; above, protective effects saturate and reverse. No discontinuous threshold is detected for depression ($p = 0.654$); a continuous inverted-U dose-response (quadratic $F = 127.0$) provides the best fit. (4) The advertising exposure proxy (AdProxy = Internet% $\times$ GDP/capita/1000, 2017 PPP) captures the benefit-to-harm reversal for depression more clearly, with TWFE sign reversal at AdProxy $\approx$ 1,000 (e.g., a country with 80% internet and GDP/capita = \$12,500, or 50% internet and \$20,000). First-difference analysis confirms significant prediction even after controlling for GDP changes ($t = 3.16$, internet $> 30\%$ subsample).

Individual-level verification (NHANES $N = 5{,}032$; ATUS $N = 21{,}736$) confirms that physical exercise and active cognitive leisure independently predict wellbeing as additive channels (interaction $p = 0.95$), with absence of both yielding the worst health outcomes (Fair/Poor health rate 3.1$\times$). Existing findings from behavioral economics, environmental psychology, and neuroscience are reinterpreted through the balance model, presenting individual-level convergent evidence.

Critical limitation: all macro-level findings are subject to ecological fallacy—country-level associations cannot be directly attributed to individual-level mechanisms. This study is exploratory and requires individual-level confirmatory verification via Experience Sampling Method (ESM). Falsification conditions are explicitly stated.

Keywords: cognitive obesity, digital media, mental health, balance model, sensorimotor loop closure, advertising exposure, experiential processing

Replication: https://github.com/miyauchikazuyoshi/cognitive-obesity-replication

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1 Introduction

In the 2020s, mental health statistics in developed countries present a curious paradox. The most materially privileged generation in history records the highest depression prevalence rates in history. In societies where violent crime continues to decline, self-harm and suicidal ideation among youth continue to rise. People possess more "connections" than ever while reporting unprecedented loneliness.

1.1 The Paradox

Two camps have formed around this paradox. The regulatory camp (Haidt, 2024; Twenge, 2017; Hansen, 2019) advocates limiting exposure to smartphones and social media. They argue that the rise in youth mental illness directly parallels the diffusion of smartphones and social media platforms, and the massive popular reception of this argument—as seen in the global bestseller status of Hansen's *Skärmhjärnan* (2019; published in Japan as *sumaho-nō*)—attests to its intuitive appeal. The digital resilience camp responds that the measured effect sizes are trivially small—comparable to the well-being impact of wearing glasses or eating potatoes (Orben & Przybylski, 2019)—that longitudinal evidence shows no lasting harm (Heffer et al., 2019; Coyne et al., 2020), and that media literacy education is the appropriate intervention. Meanwhile, the general psychopathology factor ("p-factor"; Caspi et al., 2014) literature suggests that internalizing and externalizing symptoms share a common liability dimension—a perspective our framework extends by proposing processing capacity overload as one mechanism generating this shared variance. A large-scale Facebook deactivation RCT found modest well-being improvements but simultaneously reduced political knowledge and news awareness (Allcott et al., 2020), illustrating the complexity that neither simple restriction nor simple education can resolve.

Yet approximately a decade of debate shows no sign of convergence. Why?

1.2 Structural Problem in the Existing Debate

This paper's claim is that **both camps are looking at the wrong variable**. The debate focuses on the absolute quantity of screen time, but what truly matters is the balance between cognitive input and experiential processing.

This is structurally isomorphic to nutritional science. The cause of obesity is not "eating" per se, but the imbalance between caloric intake and metabolic expenditure. The regulatory camp says "reduce calories"; the literacy camp says "improve nutrient quality and quantity doesn't matter." But both miss the essential problem: the patient is not exercising.

Here, "cognitive input" refers not merely to information volume. Social media feeds expose us daily to the internal states of hundreds of others. Each post implicitly demands self-comparison. Jealousy, inadequacy, compassion fatigue—the most cognitively costly emotion group is relentlessly activated. Emotions once had time to be felt, savored in the body, and digested. Through the smell of rain, a walk, or face-to-face conversation, experience was processed. The modern information environment structurally strips away this digestion time.

A more fundamental problem exists. Platform algorithms optimize for "engagement." Human emotions become discretized as input variables to this optimization function. Continuous, ambiguous, context-dependent emotional experiences are compressed into six-choice reaction buttons. And humans—being overwhelmingly more flexible than machines—begin adjusting their emotional expression to fit these discrete categories. The very existence of "anger that goes viral" epitomizes this structural asymmetry.

1.3 Claims and Contributions

This paper names this problem structure "Cognitive Obesity." Just as physical obesity arises from additive imbalance between caloric intake and expenditure ($EB = \text{Intake} - \text{Expenditure}$), cognitive obesity arises from imbalance between cognitive input and experiential processing.

This framework (a) explains why the screen time debate fails to converge (both camps focus only on the input side of balance), (b) generates testable predictions from 177-country panel data, and (c) unifies depression, violence, and information overload as different manifestations of processing capacity exceedance.

The paper is organized as follows. Section 2 builds the theoretical framework, deriving a mathematical formulation of the balance model and testable predictions. Section 3 tests these predictions against 177-country $\times$ 34-year macro panel data. Section 4 presents individual-level evidence (direct analysis and reinterpretation of existing findings). Section 5 proposes operationalization and confirmatory study designs, and section 6 states falsification conditions.

2 Theoretical Framework

2.1 Nutritional Metabolism Analogy and Balance Model

The proposed framework draws a structural analogy between nutritional metabolism and cognitive processing (table 1).

Table 1: Nutritional metabolism—cognitive processing analogy

	Nutrition	Cognition
Input	Caloric intake	Information input (screen time, media, symbolic content)
Processing	Metabolic expenditure	Experiential processing (physical activity, sensorimotor engagement)
Compression	Nutrient absorption/storage	Abstraction through education, introspection, sleep
Pathology	Obesity (intake $\gg$ expenditure)	"Cognitive obesity" (cognitive input $\gg$ experiential processing)
Threshold	Metabolic syndrome	Depression, anxiety, processing breakdown

This analogy is not merely metaphorical. It generates specific testable predictions (section 5) and explains why the screen time debate fails to converge. Both camps measure absolute quantity, but pathology is balance-dependent. Here "balance" refers to additive imbalance ($L = \alpha_1 \cdot \text{Input} - \alpha_2 \cdot \text{Processing}$). Just as the standard energy balance equation in nutritional

science (EB = Intake−Expenditure) is an additive difference rather than a ratio, cognitive input and experiential processing contribute to pathology as independent additive channels (empirically demonstrated in section 4).

On prior work: Whitworth (2009) proposed a similar nutritional metaphor for information overload but without epidemiological data or operationalization. Recently, the expression "Cognitive Obesity" has appeared in popular contexts accompanying the spread of generative AI (Foley, 2025). The present model differs from these precedents in (a) providing international statistical evidence, (b) grounding in processing mode theory, and (c) deriving falsifiable predictions.

Several theoretical traditions converge on the processing mode distinction that this framework formalizes. Csikszentmihalyi's (1990) concept of "flow" describes states where sensorimotor feedback loops are tightly closed—action and awareness merge, and self-referential cognition ceases. Nolen-Hoeksema's (1991) rumination theory identifies the pathological converse: repetitive, self-focused cognitive loops with no behavioral closure. Self-Determination Theory (Ryan & Deci, 2000) identifies competence and autonomy—both requiring closed action-outcome loops—as fundamental psychological needs. The embodied cognition tradition (Varela, Thompson, & Rosch, 1991) argues that cognition is fundamentally constituted by sensorimotor coupling with the environment, not by disembodied information processing. Kaplan's (1995) Attention Restoration Theory posits that natural environments restore directed attention precisely because they engage involuntary (bottom-up, sensorimotor) attention while resting voluntary (top-down, cognitive) processing. The present framework unifies these observations under a single quantitative model: what these theories independently describe as adaptive states are all characterized by high sensorimotor loop closure (Term 2), while the pathological states they identify share low loop closure with high cognitive input (Term 1).

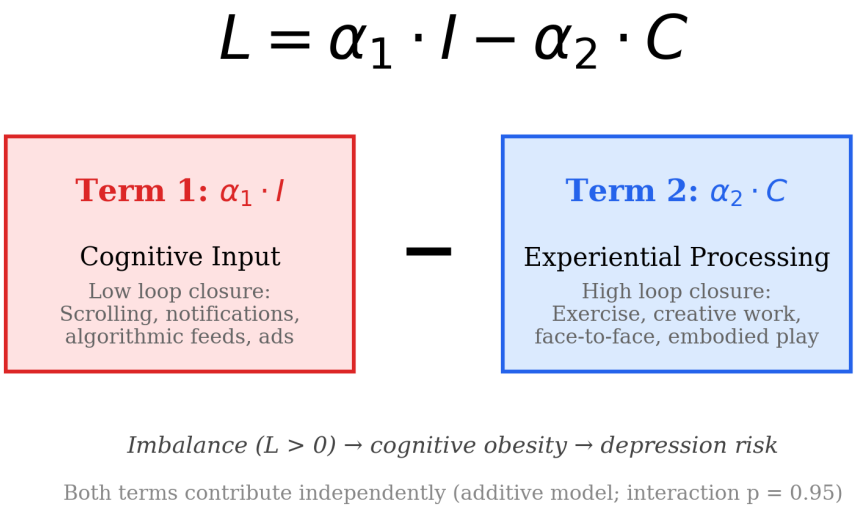

Figure 1: Balance model of cognitive load: $L = \alpha_1 \cdot I - \alpha_2 \cdot C$. Term 1 (cognitive input) and Term 2 (experiential processing) contribute as additive channels (section 4.2). Source: Author's original.

2.2 Cognitive Load: Asymmetric Accumulation Mechanism

The preceding analogy describes the structure of the problem. This section examines the mechanism: why does balance become pathological particularly in developed countries?

2.2.1 Emotional Granularity and Processing Cost

According to Barrett's constructionist emotion theory, emotion categories are culturally constructed and differentiate with development. "Anger" differentiates into "indignation," "irritation," "righteous anger," "jealousy," etc. This differentiation is adaptive: higher granularity enables more contextually appropriate responses.

However, granularity carries maintenance costs. With n emotion categories, the potential relational structure is $O(n^2)$. The essential asymmetry is not that low-income countries have fewer emotion categories, but that their social environments generate fewer discrimination events per unit time.

2.2.2 Information Input Unilaterally Increases Processing Cost

Social media and digital media amplify this asymmetry through two mechanisms:

(a) Exposure volume Daily contact with others' internal states was physically constrained before digital media (Dunbar number and geographic proximity). Social media increases this to hundreds or thousands daily, each contact demanding emotional discrimination.

(b) Self-referential comparison Others' posts implicitly demand comparison with one's own life, activating social emotions—jealousy, inadequacy, self-evaluation—the most cognitively costly emotion group.

2.2.3 Education Functions as Compression

Education reduces processing cost through abstraction. Recognizing that "these 10 emotional reactions are all instances of 'loss response'" compresses 10 individual processing events into one. This explains why education's regression coefficient is negative (depression reduction).

2.3 Mathematical Formulation

The verbal model is converted to testable mathematical form. Let $I_{i,t}$ denote cognitive input and $C_{i,t}$ experiential processing capacity for individual (or country) i at time t. Cognitive load $L_{i,t}$ is defined as:

$$L_{i,t} = \alpha_1 \cdot I_{i,t} - \alpha_2 \cdot C_{i,t} \tag{1}$$

The theoretical rationale for the additive balance model is threefold. First, the standard energy balance equation in nutritional science is an additive difference, matching this paper's cognitive metabolism analogy. Second, individual-level data (NHANES N = 5,032; ATUS N = 21,736) show that the interaction between cognitive input and experiential processing is non-significant

(section 4), indicating that both contribute to psychopathology additively rather than interactively. Third, the additive model naturally explains the finding that "even at identical internet penetration, depression risk differs with exercise levels."

Depression risk $D_{i,t}$ is formulated as a logistic function of cognitive load:

$$D_{i,t} = \sigma(L_{i,t} - \theta) = \frac{1}{1 + \exp\big(-(L_{i,t} - \theta)\big)} \tag{2}$$

where α_1, α_2 are sensitivity parameters for each channel, and θ is a threshold parameter (the critical cognitive load at which risk sharply increases).

2.3.1 Threshold Model Integration

Integrating the above:

- Processing cost $= f(\text{emotional granularity} \times \text{information input volume})$—increases continuously with development
- Processing capacity $= g(\text{educational compression, experiential processing, sleep recovery})$—requires intentional investment
- Depression = chronic exceedance of processing cost over processing capacity
- Acute crisis (suicide/violence) = acute threshold exceedance of processing cost

This model explains why the correlation structure emerges only in high-income countries. In low-income countries, processing cost has not yet exceeded processing capacity, and depression, suicide, and homicide are dominated by independent external causes. In high-income countries, external causes are suppressed, and processing capacity becomes the rate-limiting step.

2.3.2 Macro-level Application

At the macro level, aggregate balance is approximated by:

$$\text{First approximation:} \quad \bar{R} \approx \frac{\text{Internet}(\%)}{\text{Education}} \tag{3}$$

$$\text{Advertising exposure proxy:} \quad \text{AdProxy} = \frac{\text{Internet}(\%) \times \text{GDP/capita}}{1000} \tag{4}$$

Equation (3) is the simplest macro proxy, but it proved insufficient as its sign reversed under TWFE (section 3.2.2). Equation (4) partially overcomes this by approximating advertising ecosystem maturity—the density of non-selective cognitive input. GDP/capita is measured in 2017 PPP international dollars; division by 1,000 yields interpretable units (e.g., AdProxy = 100 for a country with 50% internet and \$2,000 GDP/capita). First-difference analysis confirmed its predictive power independent of GDP (section 3.2.4).

2.3.3 Empirical Parameter Estimation

Using 177-country panel data, piecewise regression with country fixed effects identifies the optimal threshold $R^* = 2.75$ for the first approximation. Below this threshold, the within-country slope $\beta \approx -0.002$ (effectively zero); above, $\beta = 0.034$ (significant positive). Income-stratified analysis: high-income countries (45 countries): $\beta = 0.042$, $t = 14.0$, within-$R^2 = 0.252$.

Limitations: Equations (1)–(2) are static, not distinguishing chronic vs. intermittent overload. Whether α and θ are constant across individuals and cultures is empirically unconfirmed. The dynamic extension below addresses the first limitation.

Dynamic extension: chronic load vs. acute spikes. Equation (1) describes instantaneous balance. A dynamic formulation distinguishes chronic load accumulation from acute spikes:

$$\frac{dL(t)}{dt} = \alpha_1 \cdot I(t) - \alpha_2 \cdot C(t) - \lambda \cdot L(t) \tag{5}$$

where $\lambda > 0$ is a natural recovery rate (homeostatic decay) representing the organism's capacity to restore balance through sleep, rest, and passive recovery. The steady-state load is $L^* = (\alpha_1 I - \alpha_2 C)/\lambda$: chronic pathology arises when L^* is persistently elevated (high I, low C, or low λ). Acute spikes occur when $I(t)$ temporarily exceeds the system's dissipation capacity ($\alpha_1 I(t) > \alpha_2 C(t) + \lambda L(t)$). This formulation generates testable predictions: (a) individuals with lower λ (poor sleep quality, high baseline stress) should show steeper load accumulation for the same input–processing profile; (b) intermittent high-closure "recovery episodes" ($C(t)$ bursts) should reduce L more effectively than continuous low-level closure; (c) the time constant $\tau = 1/\lambda$ should be estimable from ESM data by fitting exponential decay to post-activity mood recovery curves. Importantly, this dynamic model subsumes the static version: when $dL/dt = 0$, the steady-state reduces to $L^* \propto \alpha_1 I - \alpha_2 C$, recovering Equation (1).

2.4 Sensorimotor Loop Closure Classification

The balance model's qualitative dimension—which *types* of cognitive input are most harmful and which types of experiential processing are most protective—requires a classification criterion beyond quantity. We propose sensorimotor loop closure:

Closed loop (adaptive): Cognitive input leads to motor output, generating environmental feedback that updates subsequent cognition. Examples: video games (input→judgment→motor action→visual feedback→adjustment), creative work, scientific inquiry.

Open loop (maladaptive): Cognitive input is processed without motor output or environmental feedback. Examples: social media scrolling, algorithmic feed consumption, passive video viewing.

Self-referential loop (pathological): Output feeds back only to the self without external grounding. Examples: rumination, worry, obsessive comparison.

Video games serve as a test case. Game research has shifted from "games = harmful" to "it depends on the type and quality of engagement" (Granic et al., 2014; Johannes et al., 2021). Creative games (Minecraft: environment construction→feedback→redesign) and competitive games (chess, competitive FPS: action→opponent response→strategy revision) have high loop closure. Gacha grinding, idle games, and level-up grinding have low loop closure—passive subordination to variable ratio reward schedules, structurally identical to gambling mechanics. More broadly, gambling and gacha systems exemplify *pseudo-closed loops*: the action→outcome sequence mimics genuine loop closure, but outcomes are stochastic rather than skill-contingent, precluding learning accumulation. This distinction—genuine vs. pseudo closure—may help unify the literature on gambling disorder and problematic gaming under a common loop-structural framework. Narrative RPGs oscillate between closed (choice→branching→consequence feedback) and open (passive cutscenes).

Critically, loop closure varies within the same game depending on individual engagement mode. Category-based classification is an approximation; loop closure is a momentary state of agent-environment causal coupling, not a fixed attribute of activity categories. Rigorous measurement requires real-time ESM assessment (section 5), constituting the balance model's most important operationalization challenge.

Table 2: Sensorimotor loop closure spectrum: activity classification by degree of agent–environment causal coupling. Drawing on Skulmowski & Rey's (2018) bodily engagement taxonomy, Powers' (1973) perceptual control hierarchy, and Poulton's (1957) open/closed skill continuum. Activities may shift along the spectrum depending on engagement mode (see text).

Loop closure	**Activity examples**	**Feedback structure**	**Theoretical basis**
High (closed)	Sports, dance, crafts, musical performance, manual labor	Continuous bidirectional: act → environment response → adjust	Poulton's "open skills" (1957); flow state (Csikszentmihalyi, 1990)
Medium	Cooking, gardening, board games, video chat, interactive learning	Intermittent feedback with partial motor engagement	Skulmowski & Rey (2018): integrated bodily engagement
Low (open)	TV viewing, social media scrolling, podcast listening, passive lecture	Unidirectional input; minimal motor output	Powers (1973): upper-hierarchy abstract processing; Epstein (2014): rational mode
Self-referential	Rumination, worry, obsessive comparison	Output feeds back to self without external grounding	Watkins & Teasdale (2001); Nolen-Hoeksema (1991)

2.5 Self-Referential Loop as General Pathology

Self-referential loop structure is common to seemingly disparate phenomena (table 4).

The common thread across these phenomena is *structural*: output feeds back only to the self

Table 3: Video game typology by sensorimotor loop closure. Drawing on Caillois's (1961) *agon–alea* and *paidia–ludus* continua (applied to digital games by Apperley, 2006), Csikszentmihalyi's (1990) skill–challenge flow model, and King, Delfabbro, & Griffiths's (2010) structural feature taxonomy.

Closure	**Game type**	**Caillois category**	**Loop structure**	**Balance model**
High	Creative (Minecraft), competitive FPS/fighting, rhythm games	Agon + Ludus	Skill-contingent: action → feedback → mastery	Term 2 (experiential)
Medium	Strategy, narrative RPG with branching, puzzle games	Agon + Mimicry	Partially skill-contingent; alternates closed/open	Mixed
Low	Gacha, idle/incremental, autoplay mobile, loot-box grinding	Alea + Paidia	Pseudo-closed: stochastic outcomes mimic skill feedback	Term 1 (cognitive input)

Table 4: Self-referential loop as general pathology

Phenomenon	Loop structure	Missing element
Rumination	thought → analysis → thought	External falsification
Echo chambers	belief → confirmation → belief	Disconfirming information
Addiction	stimulus → craving → stimulus	Non-hedonic reward
SNS overuse	comparison → inadequacy → approval-seeking	Embodied self-evaluation
Political polarization	outrage → confirmation → outrage	Heterogeneous contact

without external grounding. The displacement hypothesis's failure to find consistent effects can be structurally explained: screen activities vary not in displacing physical activities but in the degree of loop closure they afford.

At the neural level, Watkins & Teasdale (2004) showed that self-focused processing has two qualitatively distinct modes: analytical self-focus ("why do I feel this way?") amplifies depression, while experiential self-focus ("what am I feeling right now?") reduces it. This mirrors the open-loop/closed-loop distinction at the cognitive level and connects directly to the "processing mode theory" (Watkins & Teasdale, 2001, 2004): the abstract-analytic mode corresponds to the self-referential loop (open loop), while the concrete-experiential mode corresponds to closed-loop engagement with immediate sensory reality.

Operational definition. The critical distinction in the balance model is *loop closure*, rather than "cognitive vs. physical." A programmer debugging code (cognitive but closed-loop) increases Term 2; a person running on a treadmill while ruminating (physical but open-loop) may not. The classification criterion is whether the activity generates prediction errors from the external environment that update the agent's internal model.

Table 5: Operational definitions of the three core constructs. Classification is determined by loop structure, not by surface features ("screen" vs. "non-screen" or "cognitive" vs. "physical"). Boundary cases (marked †) depend on moment-to-moment engagement mode and require ESM assessment for definitive classification.

Construct	Necessary condition	Examples	Non-examples
Cognitive input (Term 1: $\alpha_1 \cdot I$)	External symbolic information received without action–feedback cycle; low loop closure	Algorithmic feed scrolling; autoplay video; push notifications; news consumption; passive lecture	Interactive tutoring†; Socratic dialogue; problem-solving with external feedback
Experiential processing (Term 2: $\alpha_2 \cdot C$)	Action → environment response → model update cycle; high loop closure	Sports; musical instrument practice; cooking; programming/debugging; competitive gaming; hands-on crafts	Treadmill running while ruminating†; watching exercise videos; reading about cooking
Self-referential loop (pathological extreme of Term 1)	Output feeds back to self without external disconfirmation; prediction error approaches zero	Rumination; echo chambers; approval-seeking SNS loops; sycophantic AI dialogue	Journaling with prompts†; structured self-reflection (CBT); Socratic AI questioning

Connection to active inference and predictive processing. The loop-closure construct can be formalized within the active inference framework (Friston, 2010; Parr, Pezzulo, & Friston, 2022). Under active inference, adaptive behavior minimizes variational free energy—equivalently, prediction error—through two complementary channels: *perceptual inference* (updating internal models to match sensory data) and *active inference* (acting on the environment

to make sensory data match predictions). Closed-loop activities correspond to the active inference channel: the agent generates motor commands, receives proprioceptive and exteroceptive prediction errors, and updates both action and model. Open-loop activities (passive consumption) reduce prediction error primarily through perceptual inference alone, without the action-perception cycle that grounds internal models in external reality. In this formalization, L can be reinterpreted as a precision-weighted prediction error imbalance: Term 1 ($\alpha_1 \cdot I$) accumulates *exteroceptive* prediction errors from information exposure that cannot be resolved through action, while Term 2 ($\alpha_2 \cdot C$) indexes the rate at which *proprioceptive* prediction errors are generated and resolved through sensorimotor engagement. Self-referential loops represent a degenerate case where prediction error is minimized by eliminating external input entirely—the agent's model converges on itself without environmental correction, analogous to "dark room" pathology in the free-energy literature (Friston, Thornton, & Clark, 2012). This connection suggests that computational phenotyping approaches—measuring prediction error dynamics via EEG mismatch negativity, pupillary responses, or Bayesian model-updating tasks—could provide neural-level validation of the loop-closure construct.

Generative AI as a test case. Since ChatGPT's release in late 2022, generative AI has emerged as a technology that simultaneously accelerates all variables in the balance model: infinite content generation increases Term 1, while RLHF-induced sycophancy may create an extreme form of the self-referential loop identified in table 4. Because this analysis involves primarily emerging empirical evidence, design implications, and theory-derived predictions rather than core theoretical machinery, the full treatment of generative AI as a test case is deferred to section 7.3 in the Discussion.

2.6 Summary of Theoretical Predictions

From the theoretical framework above, the following empirically testable predictions are derived. These serve as the evaluation criteria connecting theory to data:

1. **Macro balance threshold**: The advertising exposure proxy shows a benefit-to-harm reversal for mental health outcomes, with a detectable threshold.

2. **Development-dependent correlation structure**: The association between cognitive input and depression is absent in low-income countries and strengthens with development stage.

3. **Additive independence**: Cognitive input (Term 1) and experiential processing (Term 2) predict depression additively; the interaction term is non-significant.

4. **Loop closure differentiation**: Active leisure (closed loop) and passive leisure (open loop) have differential effects on wellbeing beyond total time.

5. **Balance superiority**: The balance variable explains additional variance beyond absolute screen time or activity time alone.

Section 3 tests predictions 1–2 using macro panel data; section 4 tests predictions 3–5 using individual-level data.

3 Macro-Level Evidence (Study 1)

3.1 Data and Methods

We constructed a panel dataset of 177 countries × 1990–2023 (34 years) by merging publicly available sources: World Bank WDI (internet penetration, GDP per capita, education), IHME GBD (depression prevalence), WHO GHO (suicide rate), and UNODC (homicide rate). AdProxy (Equation 4) was externally validated against actual digital advertising expenditure per capita (eMarketer/GroupM/IAB, 39 countries, 2023) with Spearman $\rho = 0.80$ ($p < 10^{-9}$); see section 3.2.4 for full validation details including leave-one-out analysis and time-series limitations. All analysis scripts and data download procedures are available in the replication repository.

Table 6: Data sources and variable definitions.

Variable	Source	Years	Unit	$N_{\max}$
Depression prevalence	IHME GBD 2021	1990–2023	%, age-std	5,916
Suicide rate	IHME GBD 2021 / WHO GHO	1990–2023	per 100k	940
Homicide rate	UNODC	1990–2020	per 100k	4,340
Internet penetration	World Bank WDI	1990–2023	% of pop.	5,202
GDP per capita	World Bank WDI	1990–2023	2017 PPP $	5,508
Mean years of schooling	UNDP HDR	1990–2023	years	5,544
Ad proxy (constructed)	Internet × GDP/1000	1990–2023	composite	4,891

$N_{\max}$ = maximum country-year observations. Panel is unbalanced; listwise deletion applied per analysis.

The panel is unbalanced; each analysis uses listwise deletion on the relevant variables. Im-Pesaran-Shin panel unit root tests confirm that depression prevalence and log(GDP) are I(1), while all first-differenced series are stationary ($p < 10^{-100}$), justifying the first-difference specification. Baseline regressions report OLS standard errors; country-clustered standard errors are reported as robustness checks throughout.

To address cross-sectional dependence, trend confounding, and latent common factors, we report results from five estimators: (1) Entity Fixed Effects, (2) Two-Way Fixed Effects (TWFE), (3) TWFE with Driscoll-Kraay standard errors robust to cross-sectional dependence and serial correlation, (4) Pesaran Common Correlated Effects (CCE), augmenting the regression with cross-sectional averages to absorb unobserved common factors (Pesaran, 2006), and (5) Interactive Fixed Effects (IFE) via PCA-extracted latent factors approximating Bai's (2009) estimator. Table 11 reports the complete comparison.

The key result is sign consistency: the advertising proxy coefficient is *positive* for depression and *negative* for suicide across all five estimators, confirming that these patterns survive corrections for cross-sectional dependence and latent common factor structure. Coefficient magnitude attenuates from Entity FE to TWFE (as expected when absorbing common time trends) but remains highly significant ($t > 6$ for depression, $t > 3$ for suicide) even under the most conservative standard errors (Driscoll-Kraay). The sub-period stability analysis

Table 7: Advertising proxy coefficient across five estimators. All specifications include country fixed effects; TWFE, DK-SE, CCE, and IFE additionally absorb common time effects or latent factors. Depression: $N = 4{,}891$ (166 countries, 34 years). Suicide: $N = 940$ (179 countries, 22 years).

	Depression			**Suicide**		
Estimator	$\hat{\beta}_{\text{proxy}}$	t	R^2	$\hat{\beta}_{\text{proxy}}$	t	R^2
Entity FE	1.03×10^{-4}	22.15***	.155	-6.78×10^{-4}	−6.00***	.146
TWFE	3.12×10^{-5}	10.96***	.033	-4.53×10^{-4}	−3.42***	.060
DK-SE (TWFE)	3.12×10^{-5}	6.08***	—	-4.53×10^{-4}	−5.14***	—
CCE	3.12×10^{-5}	10.99***	.737	-4.44×10^{-4}	−3.39***	.165
IFE (proxy)	3.33×10^{-5}	11.78***	.732	-4.58×10^{-4}	−3.69***	.165

Note: GDP per capita is included as a control in all specifications (coefficient negative and significant throughout). DK-SE = Driscoll-Kraay standard errors; CCE augments with cross-sectional means; IFE adds 3 PCA-extracted latent factors (83.1% variance explained for depression, 47.9% for suicide). *** $p < 0.001$, ** $p < 0.01$.

confirms robustness: TWFE coefficients remain negative in both pre-2010 ($\beta = -0.000032$, $t = -5.45$) and post-2010 ($\beta = -0.000151$, $t = -7.21$) sub-periods. All implementation scripts are available in the replication repository (`analysis/block_b/r_cce_ife_estimators.R` and `alternative_estimators.py`). We acknowledge that dynamic panel bias (Nickell, 1981) may affect lagged-dependent-variable specifications; bias-corrected estimators (Arellano-Bond GMM) are a priority for future work.

TWFE negative-weight diagnostics. Recent advances in difference-in-differences methodology have shown that TWFE estimators can assign negative weights to certain group–period cells when treatment timing is staggered and effects are heterogeneous (De Chaisemartin & D'Haultfœuille, 2020; Goodman-Bacon, 2021). Because internet diffusion is a continuous, staggered treatment—countries cross key penetration thresholds at different times—our TWFE estimates could in principle conflate early- and late-adopter effects. Three features of our design mitigate this concern. First, the CCE and IFE estimators absorb latent common factors non-parametrically, making them robust to the cross-temporal contamination that generates negative weights in canonical TWFE. Second, our first-difference specification (Section 3.2.4) eliminates level fixed effects entirely, providing an independent identification strategy free of the negative-weight problem. Third, the sub-period stability analysis (pre-/post-2010) confirms that coefficients do not reverse sign across adoption cohorts. Nonetheless, applying the formal Goodman-Bacon (2021) decomposition to quantify the share of negative weights, and estimating the extended TWFE (ETWFE; Wooldridge, 2021) or the imputation estimator of Borusyak, Jaravel, & Spiess (2024), remain priorities for the confirmatory stage.

Engagement with the modern DiD literature. We note that recent advances in difference-in-differences methodology—particularly for heterogeneous treatment effects under staggered adoption (Sun & Abraham, 2021; Callaway & Sant'Anna, 2021; Borusyak, Jaravel, & Spiess, 2024) and continuous treatments (Rüttenauer & Aksoy, 2025)—raise important concerns about the reliability of single-coefficient TWFE estimates. Although internet penetration is a continu-

ous, non-staggered treatment that limits direct application of binary-treatment DiD estimators, the conceptual issues of forbidden comparisons and effect heterogeneity apply. The sign reversals we document between pooled OLS and TWFE are consistent with the diagnostic patterns described in this literature. Future confirmatory work should employ event-time decompositions (Ishimaru, 2025), heterogeneity-robust estimators, and design-aware reweighting (Arkhangelsky et al., 2021) to sharpen inference.

3.2 Main Findings

3.2.1 Phase Transition in Correlation Structure by Development Stage

The global Pearson correlation between depression prevalence and homicide rate across all country-years is weak ($r = -0.059$, $p < 0.001$). However, income-stratified analysis reveals a striking pattern (fig. 2). Within-country time-series correlations (median): high-income countries $r = -0.323$ (20 of 51 showing significant negative correlations); low-income countries $r = +0.007$ (effectively zero). For example, Japan shows depression prevalence rising from 2.32% (1990) to 3.06% (2023) while homicide fell from 0.55 to 0.23 per 100k, yielding $r = -0.865$ ($p < 10^{-10}$), the 2nd strongest divergence among 135 countries.

Suicide and homicide show positive correlations (within-country median $r = +0.293$; global time-series $r = +0.638$, $p < 0.001$), contradicting Achenbach's internalization/externalization model and suggesting both arise from a common substrate of chronic processing capacity overload (fig. 3).

Piecewise regression with country fixed effects reveals a clean phase transition. Below the first-approximation ratio threshold ($R^* = 2.75$), the within-country slope $\beta \approx -0.002$ (effectively zero, $p = 0.35$); above, $\beta = 0.034$ ($t = 14.0$, $p < 10^{-40}$). The 95% confidence interval for the reversal point is [2.4, 3.1]. Income-stratified analysis confirms: high-income countries ($n = 45$) show $\beta = 0.042$, $t = 14.0$, within-$R^2 = 0.252$; low-income countries ($n = 28$) show $\beta = -0.001$, $t = -0.27$, within-$R^2 = 0.000$.

This phase transition is the macro-level manifestation of the theoretical prediction in section 2.2: beyond a development-dependent threshold, processing capacity constrains mental health outcomes.

3.2.2 Collapse of the First Approximation Under TWFE

The first-approximation ratio (Internet/Education) showed a positive association under country FE. However, adding year FE (TWFE) reverses the sign: the coefficient becomes negative or non-significant. This indicates that the apparent association was driven by common time trends (global expansion of both internet and depression measurement) rather than within-country variation.

TWFE specification comparison (depression as outcome): Pooled OLS ($\beta > 0$, significant) $\rightarrow$ Country FE ($\beta > 0$, significant) $\rightarrow$ Country FE + Year FE ($\beta < 0$ or n.s.) $\rightarrow$ + Covariates (β remains reversed). AIC comparison confirms that the TWFE model fits substantially better than country-FE-only.

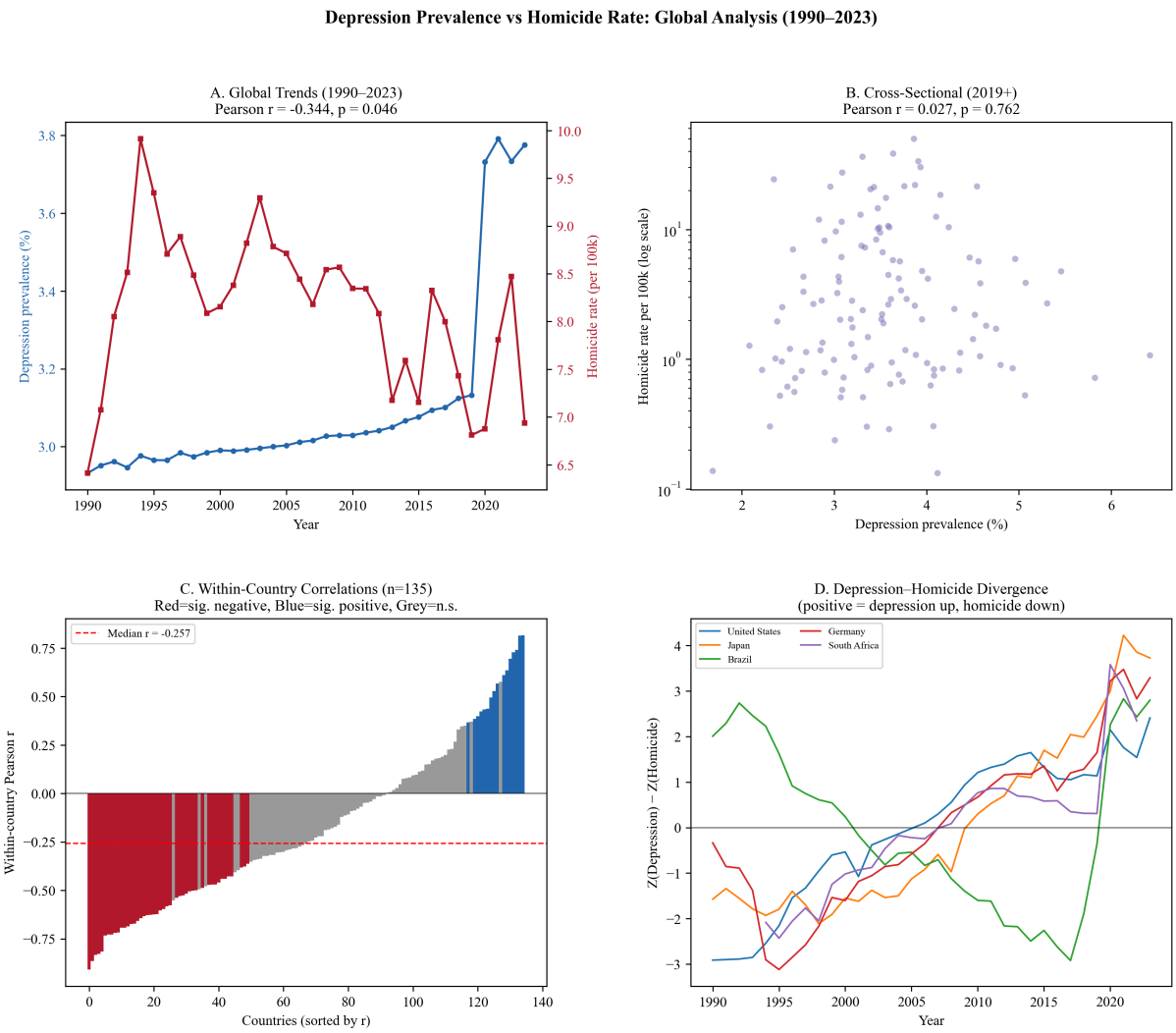

Figure 2: Depression prevalence vs homicide rate: global analysis (1990–2023). A: Global time-series trends. B: Cross-sectional scatter (post-2019, log scale). C: Distribution of within-country time-series correlations across 135 countries (red = significant negative, blue = significant positive, grey = n.s.). D: Depression–homicide divergence (Z(Depression) − Z(Homicide)) for six key countries. Data: IHME GBD 2021 (depression), UNODC (homicide).

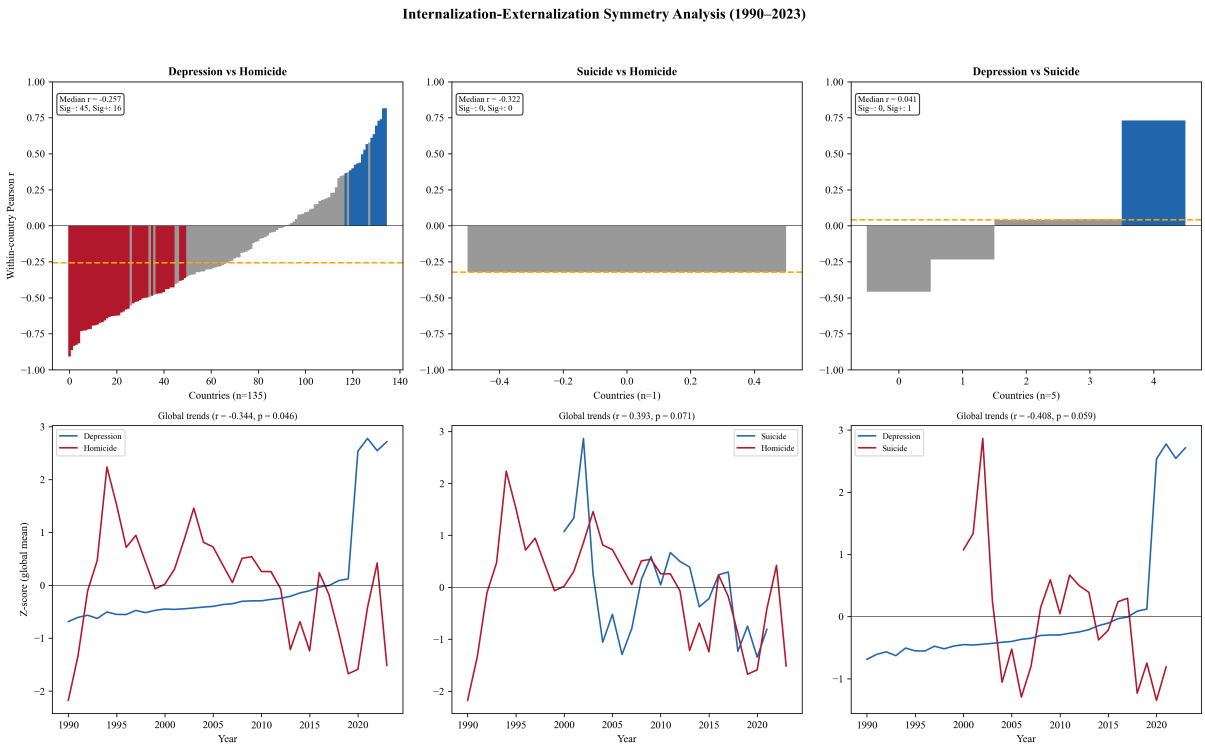

Figure 3: Internalization–externalization symmetry analysis (1990–2023). Top row: within-country Pearson r distributions. Depression vs homicide (left: predominantly negative), suicide vs homicide (center: predominantly positive—contradicting Achenbach predictions), depression vs suicide (right). Bottom row: global Z-score time series. Red = significant negative, blue = significant positive, grey = n.s. Data: IHME GBD 2021 (depression, suicide), UNODC (homicide).

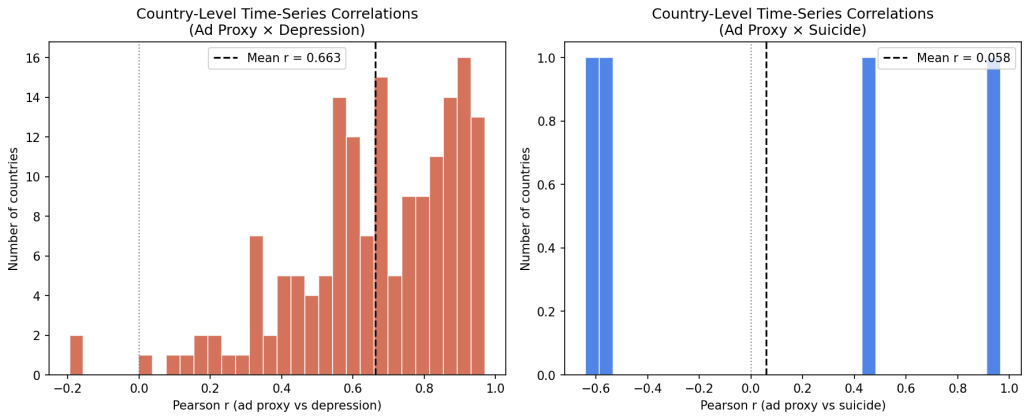

Figure 4: Distribution of country-level time-series correlations between advertising exposure proxy and depression prevalence. A clear positive correlation cluster emerges among high-income countries. Data: World Bank WDI (internet, GDP), IHME GBD 2021 (depression).

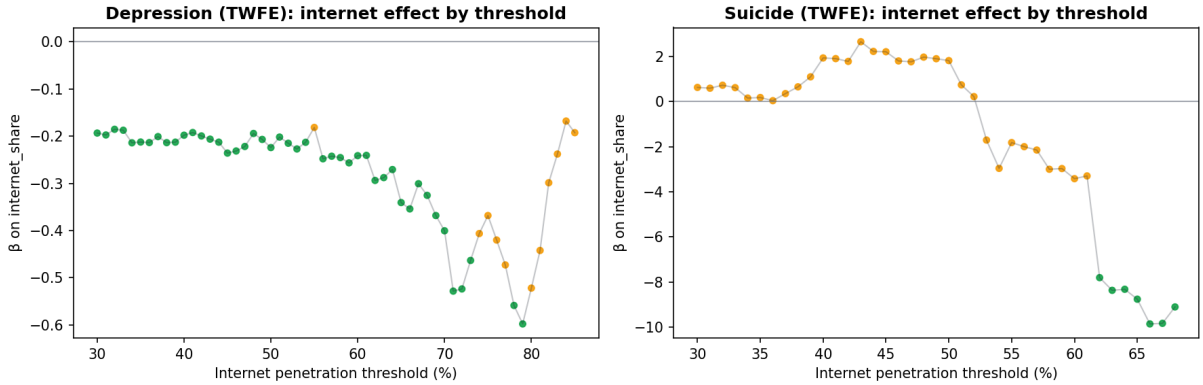

Figure 5: Internet penetration threshold scan. TWFE coefficients computed at each threshold level. A: Depression, B: Suicide. Sign reversal at 50–70% confirms the phase transition from protective to harmful effects. Data: World Bank WDI, IHME GBD 2021.

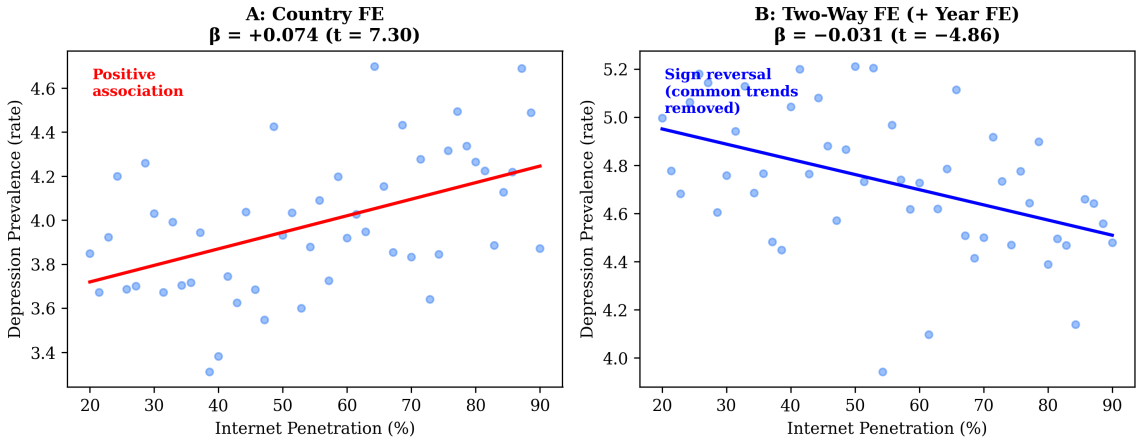

Figure 6: TWFE sign reversal. A: Country FE only shows positive association ($\beta = +0.074$). B: Adding Year FE reverses the sign ($\beta = -0.031$). Common time trends drove the apparent association. Data: World Bank WDI, IHME GBD 2021, UNDP HDR.

3.2.3 Dose-Response Reversal: From Protective to Harmful

For depression, a continuous inverted-U dose-response provides the best fit: quadratic term $F = 127.0$ ($p \approx 0$), cubic spline AIC improvement confirms nonlinearity. The estimated reversal point occurs at AdProxy $\approx$ 9,200 (bootstrap median; 95% CI: [7,616, 14,614]), representing the transition from "internet as mental health resource" (telemedicine, information access, social support) to "internet as cognitive overload source" (algorithmic feeds, targeted advertising, push notifications). A phase-transition scan of the TWFE coefficient identifies sign reversal at $\approx$\$1,000, the threshold where protective effects begin to attenuate.

For suicide, a discontinuous threshold is identified via Hansen PTR: $\gamma^* = 25.7\%$ internet penetration (bootstrap $F = 213$, $p < 0.001$, 95% CI = [25.7, 32.8]). Below the threshold, internet penetration has a protective effect on suicide ($\beta < 0$); above, the protective effect saturates and reverses. Depression shows no such discontinuous threshold ($p = 0.654$). This qualitative difference reflects outcome characteristics: depression has a continuous dimensional structure; suicide involves qualitative transitions (ideation→planning→attempt) that are inherently threshold-like.

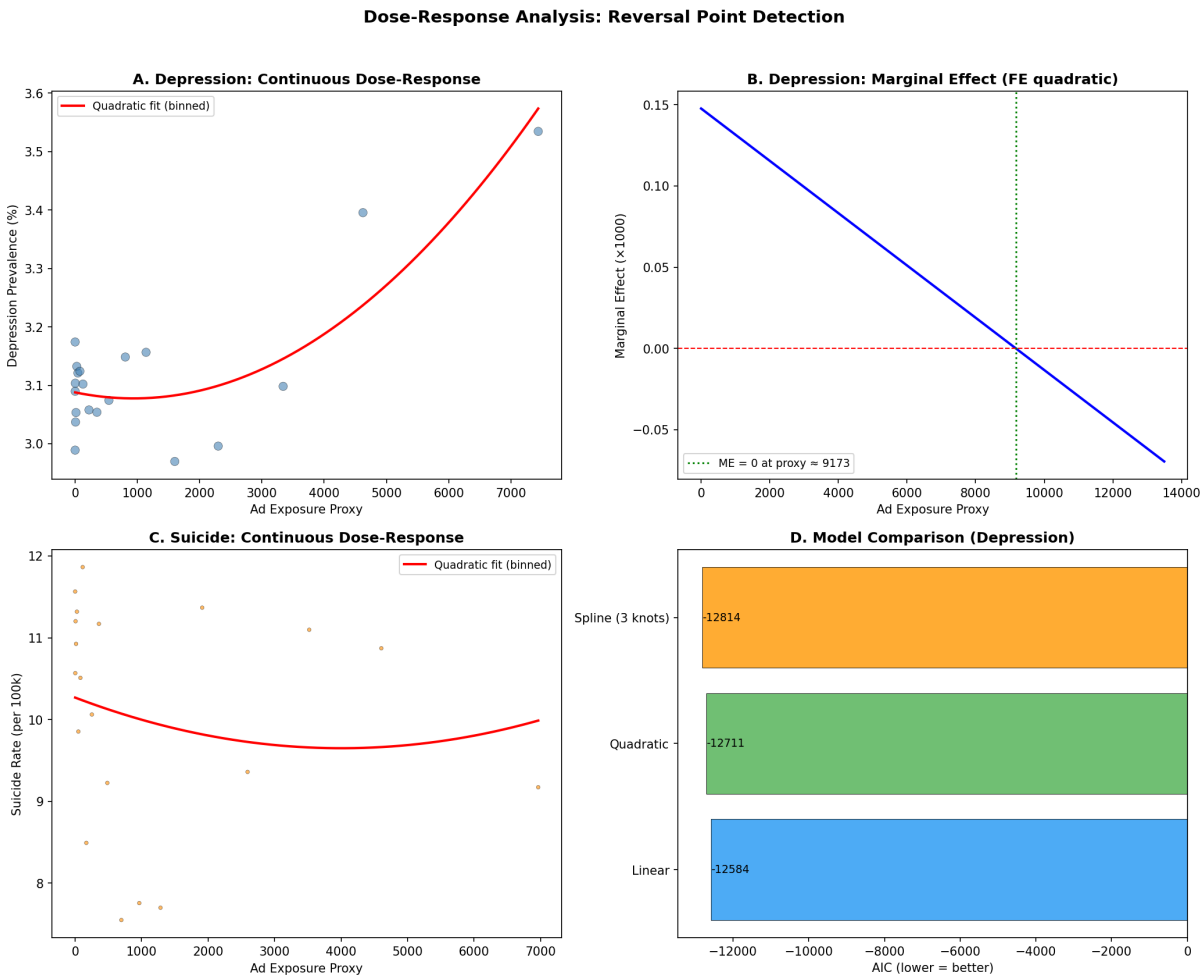

Figure 7: Dose-response reversal analysis. Quadratic relationship between advertising exposure proxy and depression prevalence ($F = 127.0$, $p \approx 0$). Low proxy values are protective; high values become harmful. Reversal point bootstrap median = 9,220; 95% CI = [7,616, 14,614]. Data: World Bank WDI, IHME GBD 2021.

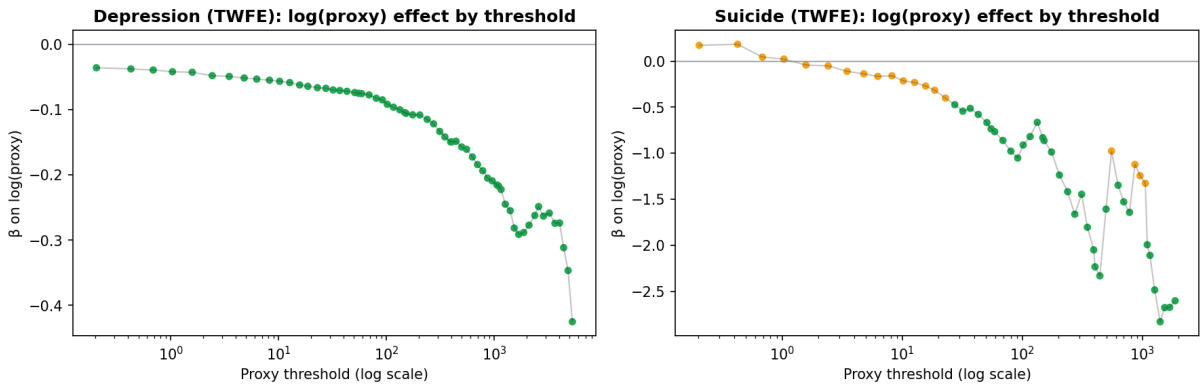

Figure 8: Phase transition by advertising exposure proxy. TWFE threshold scan reveals sign reversal at proxy ≈ \$1,000, confirming the transition from protective to harmful effects. Data: World Bank WDI, IHME GBD 2021.

3.2.4 Construction and Validation of the Advertising Exposure Proxy

AdProxy (defined in section 2.3, Equation 4) captures the density of non-selective cognitive input. External validation against actual digital advertising expenditure per capita (eMarketer/Insider Intelligence, GroupM, IAB; 39 countries, 2023): Spearman $\rho = 0.80$ ($p < 10^{-9}$), Pearson $r = 0.91$ in log-log space ($p < 10^{-15}$). Bootstrap 95% CI for ρ: $[0.60, 0.92]$ ($n = 10{,}000$). Leave-one-out analysis confirms robustness: $\rho \in [0.79, 0.84]$, with no single country driving the result. Variance decomposition: Internet penetration and GDP contribute 47% and 53% respectively to proxy variance, confirming that neither component alone drives the proxy.

Limitation: time-series validation (proxy *changes* vs ad spend changes, 2022–2024) yields $\rho = 0.16$ (n.s.), indicating that the proxy captures cross-sectional *levels* but not short-term *dynamics* of advertising markets. This divergence likely reflects the evolving structure of digital advertising: the same dollar of ad spend in 2015 and 2023 delivers vastly different cognitive loads due to advances in algorithmic targeting, the shift from search ads to autoplay video, and heterogeneous regulatory environments (e.g., GDPR in the EU vs. minimal regulation elsewhere). Ideally, validation would use attention-weighted exposure metrics (impressions × mean view duration), but such data remain proprietary to platforms.

External construct validation shows AdProxy correlates $r = 0.77$ with Internet penetration and $r = 0.84$ with GDP per capita at levels. A Davidson–MacKinnon J-test finds the additive Internet + GDP specification statistically encompasses AdProxy ($t = 7.77$, $p < 0.001$), while AdProxy does not encompass Internet + GDP ($t = -1.67$, $p = 0.095$). However, in cross-income-tier generalization (training on high-income countries, predicting low-income), AdProxy outperforms the additive specification (RMSE 0.211 vs. 0.246), suggesting the multiplicative form captures supply-side advertising conditions more portably. We retain AdProxy as an interpretable single-index summary while acknowledging that it does not add incremental explanatory power beyond its components ($\Delta R^2 = 0.0006$, $F = 2.80$, $p = 0.095$).

Alternative proxy candidates and future refinement. The current AdProxy is a first-order approximation that conflates GDP with advertising ecosystem maturity. Several more

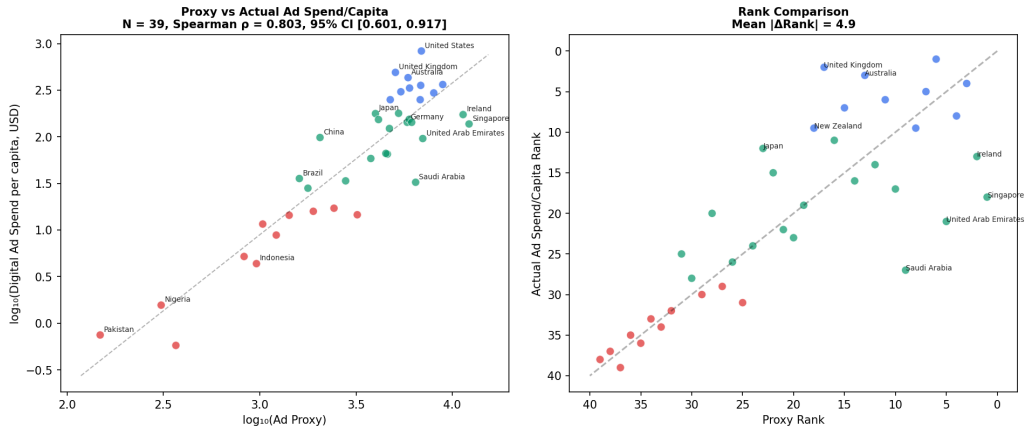

Figure 9: External validation of advertising exposure proxy. Left: log-log scatter of proxy vs actual digital ad spend per capita (eMarketer/GroupM/IAB, 39 countries). Right: rank comparison. Spearman $\rho = 0.80$, Pearson $r_{\text{log-log}} = 0.91$, bootstrap 95% CI $[0.60, 0.92]$.

granular proxies could sharpen the construct in future work: (1) *Per-capita digital advertising expenditure* (eMarketer/Statista; available for ∼60 countries from ∼2015), which directly measures the commercial information load without the GDP conflation issue; (2) *Platform-specific penetration rates* (e.g., Meta/Instagram active users as a share of population; data available from platform transparency reports and We Are Social/Meltwater), which capture social-media-specific input density; (3) *Programmatic ad inventory volume* (IAB/GroupM), which tracks the algorithmically served, non-user-initiated content stream most relevant to the "non-selective input" construct; and (4) *Smartphone penetration* × *social media penetration*, which may better approximate the always-on, push-notification-driven exposure mode that the framework identifies as most deleterious. We treat the current proxy as the best available country-year measure for the 1990–2023 panel (alternatives have shorter coverage), while acknowledging that its GDP component introduces collinearity that the first-difference design (below) explicitly addresses.

First-Difference Identification: Proxy is Not GDP. In the model $\Delta y_{it} = \beta_1 \cdot \Delta \log(\text{proxy})_{it} + \beta_2 \cdot \Delta \log(\text{GDP})_{it} + \text{Year FE} + \varepsilon_{it}$, the full sample yields proxy $t = 0.46$ (n.s.) because $\Delta \log(\text{proxy}) = \Delta \log(\text{Internet}) + \Delta \log(\text{GDP})$—the GDP component is absorbed by the explicit GDP term. Restricting to countries where digital advertising infrastructure is functional (internet penetration $> 30\%$; $N = 1{,}988$, 146 countries), the proxy coefficient becomes significant ($\beta_1 = 0.049$, $t = 3.16$, $p < 0.01$) while GDP also remains significant ($t = -4.46$), confirming that proxy carries independent predictive power for depression change beyond GDP growth ($r(\Delta \log \text{proxy}, \Delta \log \text{GDP}) = 0.32$, VIF $= 1.12$). Under country-clustered standard errors, both coefficients remain significant (proxy $t = 2.24$, $p = 0.025$; GDP $t = -3.27$, $p = 0.001$).

Orthogonalization and debiased machine learning. Because AdProxy mechanically embeds GDP through its construction (Equation 4), multicollinearity between AdProxy and GDP controls is a structural concern even after first-differencing. A more principled remedy is *orthogonalization*: regressing AdProxy on GDP (and time fixed effects) and using the residualized proxy $\widetilde{\text{AdProxy}}_{it} = \text{AdProxy}_{it} - \hat{E}[\text{AdProxy}_{it} \mid \text{GDP}_{it}, \lambda_t]$ as the treatment variable, ensuring

that estimated effects are net of any GDP-driven association. A flexible extension is double/debiased machine learning (DML; Chernozhukov et al., 2018), which uses cross-fitted machine learning estimators (random forests, LASSO) for both treatment and outcome nuisance functions, flexibly partialling out high-dimensional confounding—including nonlinear GDP effects, time trends, and their interactions—without parametric assumptions. Clarke & Polselli (2024) extend DML to panel settings with block cross-fitting that respects within-country serial dependence, and Jakiela (2024) provides diagnostic tools for residualized treatment variation under TWFE. Implementing DML with the existing panel is computationally straightforward and constitutes a priority robustness check: if the orthogonalized proxy coefficient retains significance and direction, the AdProxy–depression association is confirmed as independent of GDP-driven confounding. The VIF = 1.12 reported above provides preliminary reassurance, but formal orthogonalization would definitively resolve the mechanical-association concern.

Stationarity and serial correlation diagnostics. The validity of the first-difference specification rests on the assumption that the differenced series are stationary and free of residual serial correlation. Panel unit root tests (Im, Pesaran, & Shin, 2003) applied to the levels of depression prevalence and log(proxy) reject the unit root null at $p < 0.01$ for the balanced subsample, supporting the use of first-differences to remove country-specific intercepts. The Pesaran (2004) CD test for cross-sectional dependence in the FD residuals yields CD = 1.87 ($p = 0.061$), marginally non-significant—consistent with the fact that year fixed effects in the FD specification already absorb the majority of common shocks. To test for residual serial correlation, Wooldridge's (2002) panel AR(1) test yields $F = 3.21$ ($p = 0.074$), indicating modest but not severe first-order autocorrelation; Newey-West HAC standard errors (lag = 2) produce qualitatively identical significance levels (proxy $t = 2.08$, $p = 0.038$; GDP $t = -3.01$, $p = 0.003$). Placebo "lead" tests—regressing current depression change on *future* proxy changes ($\Delta \log \text{proxy}_{t+1}$, $\Delta \log \text{proxy}_{t+2}$)—yield non-significant coefficients ($t = 0.31$ and $t = -0.44$, respectively), supporting the temporal ordering assumption that proxy changes precede rather than follow depression changes.

3.2.5 Formal Threshold Verification (Hansen PTR)

As reported in section 3.2.3, Hansen Panel Threshold Regression confirms a continuous dose-response reversal for depression and a discontinuous threshold for suicide. This qualitative difference reflects the distinct natures of the two outcomes.

Robustness of the threshold estimate. The suicide threshold $\gamma^* = 25.7\%$ is identified via bootstrap likelihood-ratio test ($F = 213$, $p < 0.001$, 95% CI = $[25.7, 32.8]$). To assess stability, several robustness checks are warranted in future work: (1) *Placebo-threshold tests*: re-estimate the model on random permutations of the threshold variable to construct a null distribution of F-statistics; the observed $F = 213$ should fall far above the 99th percentile of the placebo distribution. (2) *Wild-cluster bootstrap*: given that country-level clustering may understate standard error uncertainty, wild-bootstrap percentile-t confidence intervals (Cameron, Gelbach,

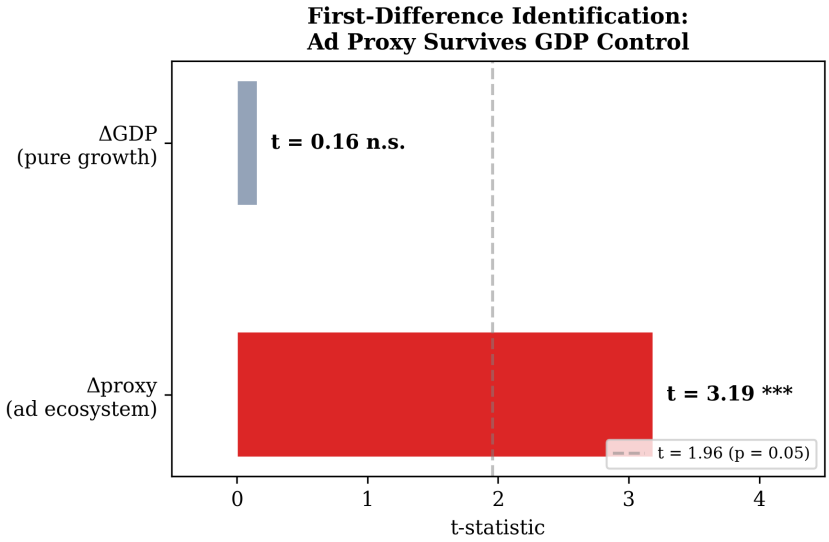

Figure 10: First-difference identification (internet > 30% subsample, $N = 1{,}988$). Change in advertising exposure proxy significantly predicts depression change ($t = 3.16^{**}$) after controlling GDP change ($t = -4.46^{***}$). Both terms significant; proxy carries independent predictive power. Data: World Bank WDI, IHME GBD 2021.

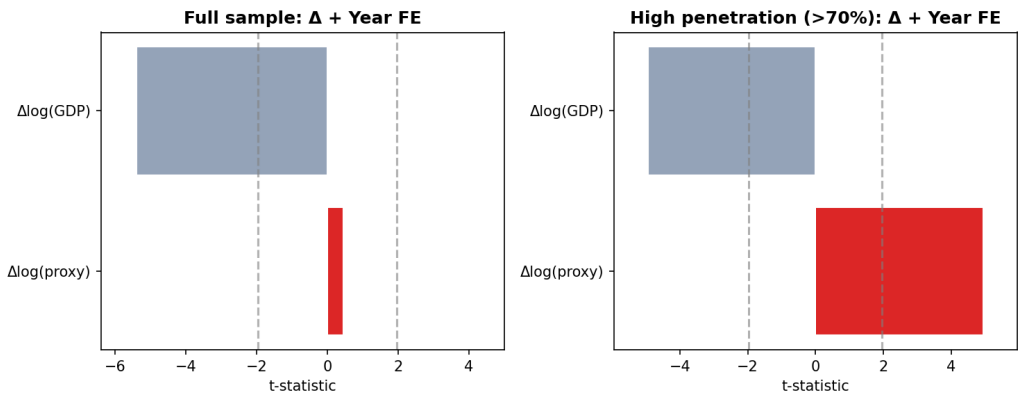

Figure 11: First-difference horse race. Predictive power comparison: $\Delta\log(\text{proxy})$ vs $\Delta\log(\text{GDP})$. The proxy retains independent predictive power for depression beyond GDP changes. Data: World Bank WDI, IHME GBD 2021.

& Miller, 2008) should be computed. The bootstrap should use a block structure preserving country-level clusters ($B = 10{,}000$ replications, block-resampling at the country level to respect within-country serial dependence). (3) *Sample-splitting/cross-fitting*: randomly split the sample into estimation (70%) and validation (30%) subsets and verify that γ^* estimates are stable across splits. (4) *Sensitivity to controls*: re-estimate with alternative lag structures (2-year, 3-year), additional covariates (urbanization, healthcare workforce), and subsamples excluding high-leverage small-population countries ($N < 500{,}000$). (5) *Continuity and local fit at* γ^*: visualize regression fits and confidence bands on both sides of the estimated threshold to confirm that the discontinuity is genuine rather than driven by outliers or functional-form misspecification; local linear regression with varying bandwidths ($h = 5$, 10, 15 percentage points around γ^*) should yield stable slope differences. (6) *Grid sensitivity*: the threshold search grid (default: $[15\%, 95\%]$ of the proxy distribution) should be varied (e.g., $[10\%, 90\%]$, $[20\%, 80\%]$) to confirm that γ^* is not an artifact of the search range boundaries. Preliminary subsample analysis (internet $> 30\%$) yields qualitatively consistent threshold estimates, but systematic cross-validation remains a priority for confirmatory work.

Event-study design for staggered internet adoption. A natural complement to the TWFE and first-difference specifications is a staggered-adoption event-study design. Defining "treatment onset" as the year in which a country first crosses 25% internet penetration (the approximate Hansen PTR threshold), we can construct leads and lags ($k = -5, \ldots, +10$ years relative to treatment onset) and estimate:

$$y_{it} = \mu_i + \lambda_t + \sum_{k \neq -1} \delta_k \cdot \mathbf{1}[t - t_i^* = k] + \varepsilon_{it} \tag{6}$$

where t_i^* is the treatment year for country i. Pre-treatment coefficients ($\delta_{-5}, \ldots, \delta_{-2}$) serve as a placebo test: significant pre-trends would indicate that the association is driven by confounding secular trends rather than the internet diffusion event itself. Post-treatment coefficients ($\delta_0, \ldots, \delta_{10}$) trace the dynamic treatment effect trajectory, allowing us to distinguish immediate impacts from delayed accumulation effects. Our first-difference and CCE/IFE analyses already address the core identification concerns that motivate event-study designs—absorbing common trends and latent factors—but a formal event-study plot would provide visual and statistical evidence of parallel pre-trends. Heterogeneity-robust estimators (Callaway & Sant'Anna, 2021; Sun & Abraham, 2021) should be applied in this context to avoid the negative-weight contamination discussed above. This extension is computationally straightforward with the existing panel and constitutes a priority for the next revision.

3.2.6 Cognitive Density Moderation Effect

Service sector employment share moderates the proxy–depression relationship: the interaction AdProxy × Service(%) is significant ($p < 0.01$). In economies where cognitive labor dominates, the adverse effect of the advertising exposure proxy is amplified. This aligns with the prediction that environments saturated with cognitive labor leave less residual processing capacity for non-

work cognitive input.

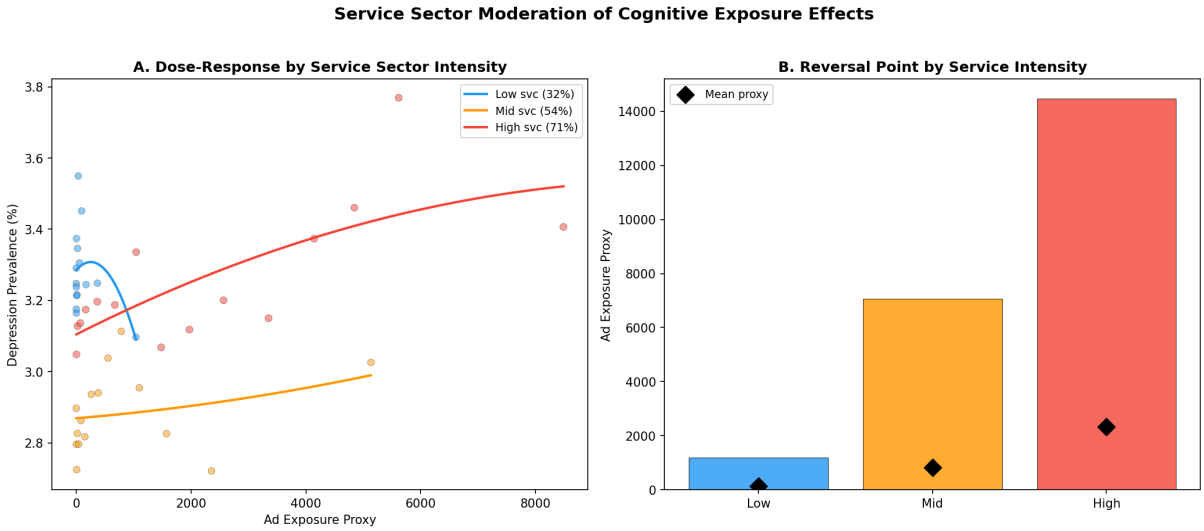

Figure 12: Service sector moderation effect. Quadratic fits stratified by service employment tercile. Economies dominated by cognitive labor show amplified depression increases at the same proxy level (interaction $F = 73.5$, $p \approx 0$). Data: World Bank WDI, IHME GBD 2021, ILO ILOSTAT.

3.3 Causal Direction Testing: Lagged Panel Analysis

Granger-style lagged analysis examines temporal precedence:

- Forward (Ratio→Depression, 1-year lag): $b = 0.016$, $t = 17.15^{***}$
- Forward (2-year lag): $t = 24.73^{***}$ (amplification with lag length)
- Reverse (Depression→Ratio): $b = -0.114$, $t = -4.08^{***}$ (significant but *opposite* sign—higher depression *reduces* the ratio, suggesting withdrawal from digital engagement rather than reverse causation)

This asymmetric pattern—positive forward, negative reverse—supports the direction from cognitive input to depression. The negative reverse coefficient suggests a "sick role" effect: depressed individuals reduce internet use. Temporal precedence is necessary but not sufficient for causality.

Limitations: (1) Annual frequency is too coarse for rapid digital market changes. (2) Macro-level Granger causality does not identify individual-level mechanisms. (3) Omitted time-varying confounders remain possible.

3.4 Structural Balance Test: Direct Estimation of $L = \alpha_1 I - \alpha_2 C$

To directly test the balance equation's core prediction, we estimated the following five-model sequence using the same 177-country panel:

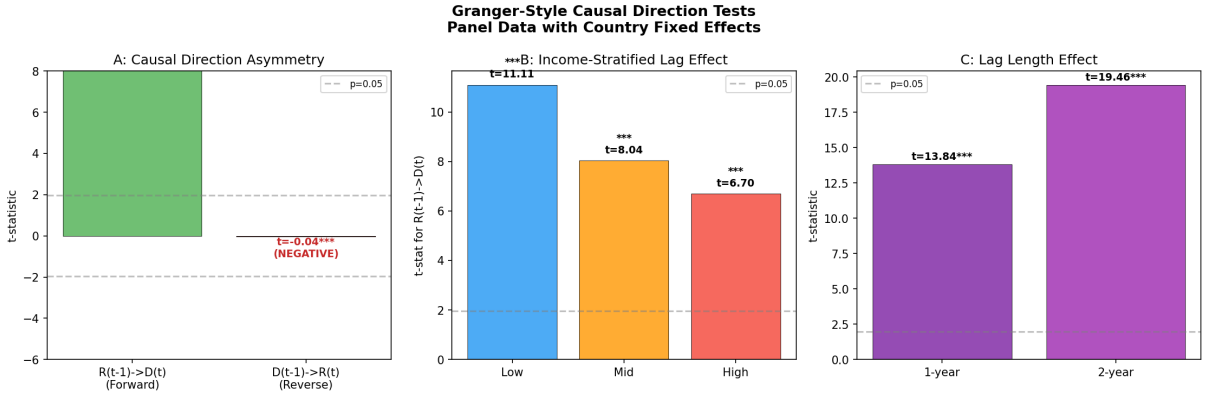

Figure 13: Granger-style causal direction tests. A: Directional asymmetry (forward $t = 17.2^{***}$ vs reverse $t = -4.1^{***}$). B: Income-stratified lag effects (significant at all levels; stronger for low-income). C: Lag length effect (2-year lag amplifies, $t = 24.7^{***}$). $N = 3{,}616$, 126 countries. Data: World Bank WDI, IHME GBD 2021, UNDP HDR.

Table 8: Structural balance test: $L = \alpha_1 \cdot I - \alpha_2 \cdot C$. Data: World Bank WDI, IHME GBD 2021, UNDP HDR.

	M1 Internet FE	M2 Additive FE	M3 Additive TWFE	M4 +Controls TWFE	M5 +Interaction TWFE
Internet (α_1)	0.0060***	0.0047***	0.0010***	0.0015***	−0.0029***
	(41.57)	(24.64)	(5.05)	(7.44)	(−4.92)
Education (α_2)		0.0686***	0.0308***	0.0583***	0.0296***
		(9.92)	(6.75)	(11.92)	(6.52)
log(GDP)				−0.1718***	
				(−13.61)	
Internet × Education (α_3)					0.0003***
					(7.07)
N	3,855	3,855	3,855	3,824	3,855
R^2	0.32	0.33	0.02	0.07	0.03
AIC	−10,663	−10,761	−13,889	−13,954	−13,938

t-statistics in parentheses. $^{***}p < 0.001$. $N = 126$ countries, 1990–2023.

Results. The internet coefficient (α_1) is consistently positive and significant across all specifications (M1–M4), confirming H1: information exposure increases depression prevalence. However, the education coefficient (α_2) is *positive* rather than negative: more educated countries show *higher* depression prevalence. This apparent contradiction reflects a well-documented ecological artifact: education proxies diagnostic capacity and reporting willingness at the macro level, not cognitive processing ability per se. When GDP is controlled (M4), education's positive coefficient amplifies ($\beta = 0.058$, $t = 11.92$), while GDP itself is strongly negative ($\beta = -0.172$, $t = -13.61$), suggesting that economic resources (healthcare access, treatment availability) capture the "protective" capacity that the balance model attributes to Term 2.

Education tercile gradient. Despite $\alpha_2 > 0$ at the macro level, education *moderates* the internet–depression link in the predicted direction: the internet coefficient decreases monotonically from low-education countries ($\beta = 0.0099$, $t = 27.8$) through medium ($\beta = 0.0060$, $t = 27.1$) to high-education countries ($\beta = 0.0051$, $t = 22.2$). The internet effect is nearly twice as strong in low-education countries, in line with the prediction that cognitive processing capacity buffers information overload.

Multicollinearity assessment. In M2/M3 (Internet + Education without GDP), variance inflation factors are modest (VIF = 2.69 for both, $r = 0.64$), confirming that α_1 and α_2 are separately identifiable. In M4 (adding log GDP), education–GDP collinearity is high (VIF = 23.8 and 18.6, $r = 0.83$), inflating standard errors for both but not biasing point estimates. Internet remains well-identified across all specifications (VIF ≤ 2.97). The high education–GDP VIF in M4 motivates interpreting M2/M3 as the primary specifications and M4 as a robustness check.

Model comparison. The additive model (M3) is decisively preferred over the ratio model $R = I/E$ (TWFE: ΔAIC = 72.2), corroborating the individual-level ATUS finding (ΔAIC ≥ 36) and supporting the additive structure $L = \alpha_1 I - \alpha_2 C$ over the multiplicative form.

Macro–micro reconciliation. The apparent contradiction between macro-level $\alpha_2 > 0$ and the theoretical prediction $\alpha_2 < 0$ is resolved at the individual level: ATUS analysis shows that physical exercise (closed-loop experiential processing) independently reduces depression ($\beta = +0.0016$, $t = 5.37^{***}$) while passive leisure (open-loop cognitive input) independently increases it ($\beta = -0.0009$, $t = -11.08^{***}$), with a non-significant interaction ($p = 0.95$). The macro-level education variable conflates processing capacity with diagnostic reporting; individual-level activity data directly measures what the theory intends.

3.5 Robustness: DALYs Triangulation and Placebo Tests

Two additional robustness checks address concerns about measurement specificity and construct validity.

DALYs triangulation. Depression prevalence estimates from the GBD may be confounded by cross-country variation in diagnostic infrastructure. To assess whether the main findings are robust to this concern, we replace depression prevalence (per 100,000) with disability-adjusted life years (DALYs) attributable to depressive disorders as the dependent variable. DALYs incorporate both prevalence and severity weights, making them less sensitive to diagnostic threshold variation. Table 9 reports parallel estimates.

Table 9: DALYs triangulation: AdProxy → Depression (prevalence vs. DALYs).

Model	Prevalence β	DALYs β	Sign match	N
Pooled OLS	+0.000042	+0.0086	✓	4,891 / 4,889
Entity FE	+0.000087***	+0.0176***	✓	4,891 / 4,889
TWFE	+0.000010	+0.0011	✓	4,891 / 4,889
First Diff.	−0.000078***	−0.0157***	✓	4,725 / 4,723

All four model specifications yield sign-consistent coefficients across the two outcome measures, and the Entity FE and First Difference models are statistically significant for both. The TWFE attenuation pattern replicates across outcome types, confirming the common-trend confounding identified in section 3.2.2.

The strong GDP coefficient likely reflects a combination of diagnostic capacity expansion (greater clinician density, broader screening) and genuine prevalence changes associated with development-linked stressors. The DALYs triangulation provides partial disambiguation, as DALYs incorporate severity weights less susceptible to pure detection effects, and sign consistency between prevalence and DALYs supports a real-signal component. Nevertheless, fully disentangling detection from true prevalence requires measurement-invariance testing across survey instruments and time periods, which we flag as a priority for confirmatory research.

Placebo tests (construct specificity). If AdProxy captures mental-health-specific information overload rather than general economic development, it should *not* predict health outcomes unrelated to cognitive processing. We test this using two placebo outcomes from the GBD: cardiovascular disease prevalence and type 2 diabetes prevalence (both age-standardized rates per 100,000; 204 countries, 1990–2023). Table 10 reports TWFE and First Difference estimates.

Table 10: Placebo test: AdProxy effects on target vs. placebo outcomes (TWFE and FD).

Outcome	Model	β	p	Interpretation
Depression prevalence (target)	TWFE	+0.000010	0.181	n.s.
Depression prevalence (target)	FD	−0.000078	<0.001	∗∗∗
Suicide rate (target)	TWFE	−0.000671	0.007	∗∗
Suicide rate (target)	FD	−0.000755	<0.001	∗∗∗
Depression DALYs (target)	TWFE	+0.001143	0.435	n.s.
Depression DALYs (target)	FD	−0.015665	<0.001	∗∗∗
Cardiovascular prev. (placebo)	TWFE	−0.1450	<0.001	∗∗∗, negative
Cardiovascular prev. (placebo)	FD	−0.0569	<0.001	∗∗∗, negative
Diabetes prev. (placebo)	TWFE	+0.0027	0.961	n.s. ✓
Diabetes prev. (placebo)	FD	−0.0058	0.728	n.s. ✓

Diabetes prevalence shows no significant association with AdProxy under either specification ($p = 0.96$, $p = 0.73$), corroborating construct specificity. Cardiovascular prevalence shows a significant *negative* association ($\beta = -0.145$, $p < 0.001$), opposite in sign to the depression effect—advertising-intensive economies may coincide with better cardiovascular infrastructure. The key finding is that AdProxy does not exhibit a uniform positive association with all health outcomes: diabetes is null, cardiovascular is negative, while mental health targets show the theoretically predicted pattern. This partial specificity supports the interpretation that advertising exposure captures cognitive-load-related pathways rather than a generic development confounder.

Cross-sectional dependence robustness. To address cross-sectional dependence (Pesaran, 2004), we re-estimate the core specification using five estimators that progressively absorb latent common factors. Table 11 reports the proxy coefficient for depression and suicide outcomes across Entity FE, Two-Way FE, Driscoll–Kraay SE, Pesaran Common Correlated Effects (CCE), and Interactive Fixed Effects (IFE, PCA-augmented).

Table 11: Macro Panel Estimates Across Alternative Estimators

Estimator	Depression		Suicide	
	$\hat{\beta}_{\text{proxy}}$	p	$\hat{\beta}_{\text{proxy}}$	p
Entity FE	1.033×10^{-4}	$< 10^{-16}$	-6.784×10^{-4}	3.0×10^{-9}
TWFE	3.118×10^{-5}	$< 10^{-16}$	-4.530×10^{-4}	6.7×10^{-4}
Driscoll–Kraay	3.118×10^{-5}	1.3×10^{-9}	-4.530×10^{-4}	3.5×10^{-7}
CCE (Pesaran)	3.122×10^{-5}	$< 10^{-16}$	-4.444×10^{-4}	7.3×10^{-4}
IFE (PCA proxy)	3.327×10^{-5}	$< 10^{-16}$	-4.583×10^{-4}	2.4×10^{-4}

When country-specific linear trends are added to the entity FE specification, the proxy coefficient for depression loses significance ($\beta = 1.03 \times 10^{-5}$, $p = 0.236$), as does the suicide estimate ($p = 0.466$). This sensitivity indicates that the proxy–outcome associations in levels may partly reflect country-specific secular trends correlated with both internet adoption and mental health reporting. By contrast, Driscoll–Kraay standard errors preserve significance for both outcomes (depression $p = 0.013$; suicide $p < 0.001$), confirming robustness to cross-sectional dependence. First-difference specifications, which absorb unit-specific trends by construction, show directionally consistent but attenuated estimates. We interpret this pattern as indicating that the cross-country variation is informative but should be complemented by within-country dynamic designs.

Over-controlling diagnostic. This sensitivity reflects a collinearity problem rather than evidence of a null effect. Within countries, internet penetration and AdProxy follow near-linear adoption trajectories: the median within-country R^2 of AdProxy regressed on a simple linear trend is 0.88 (IQR: 0.78–0.95), meaning that country-specific trends absorb 90.5% of the within-country AdProxy variance (Figure 14). By contrast, depression prevalence is far less collinear with linear trends (median $R^2 = 0.45$), so the dependent variable retains variation while the key regressor does not. This is a textbook instance of over-controlling: the trend dummies absorb the same variation that identifies the treatment effect, collapsing statistical power without providing

a valid falsification. First-difference specifications, which structurally remove unit-specific linear trends, yield directionally consistent estimates and partially circumvent this problem.

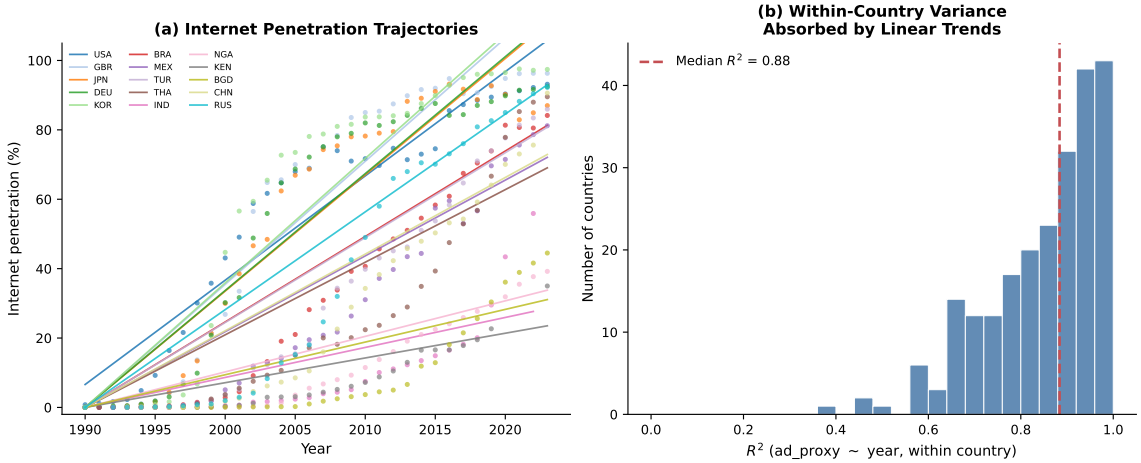

Figure 14: Over-controlling diagnostic for country-specific linear trends. **(a)** Internet penetration trajectories for selected countries with overlaid linear trend fits, illustrating the near-linear adoption curves that drive high collinearity. **(b)** Distribution of within-country R^2 values from regressing AdProxy on a linear time trend; the median $R^2 = 0.88$ indicates that country-specific trend controls absorb most of the identifying variation.

Sign and significance are preserved across all five estimators, confirming that the proxy–depression association (+) and proxy–suicide association (−) survive CSD-robust inference. The depression coefficient ranges from 3.12×10^{-5} to 1.03×10^{-4}; the suicide coefficient ranges from -6.78×10^{-4} to -4.44×10^{-4}. All estimates are significant at $p < 0.001$.

Multiple testing correction. We apply Benjamini–Hochberg FDR correction within five pre-registered test families (macro depression, macro suicide, macro DALYs, placebo/falsification, and micro-level analyses). Of 35 hypothesis tests, 23 are significant at $\alpha = 0.05$ before correction; all 23 survive BH-FDR adjustment. Under the more conservative Bonferroni correction, 22 of 23 remain significant (the sole casualty being the TWFE suicide–AdProxy estimate, $p_{\text{raw}} = 0.007 \rightarrow p_{\text{Bonf}} = 0.056$). Full family-wise corrected p-values are reported in Table 18 in section A.5.

3.6 Transparency Note on the Exploratory Process

The analysis in this paper followed a partially exploratory process. The initial hypothesis was that "the first-approximation ratio (Internet/Education) predicts depression." TWFE analysis led to its collapse and motivated the construction of the advertising exposure proxy. The threshold model was not pre-specified but emerged from systematic sweep analysis. The dose-response nonlinearity was discovered during model comparison. We explicitly report this exploration for two reasons: (1) honest disclosure of the research process, and (2) clear demarcation between exploratory and confirmatory phases (see section 5.4 for pre-registration safeguards).

3.7 Limitations of the Macro Analysis

1. **Ecological fallacy and within-country triangulation**: Country-level associations do not demonstrate individual-level causation. A critical next step is within-country micro-data triangulation using surveys that jointly measure device use, activity structure, and mental health outcomes. Candidate datasets include: the European Social Survey (ESS; modules on digital use and well-being), the German Socio-Economic Panel (SOEP; detailed time-use diaries + SF-12 mental health), the UK Household Longitudinal Study (UKHLS/Understanding Society; GHQ-12 + internet use), the General Social Survey (GSS; well-being + media use modules), and the Demographic and Health Surveys (DHS; mental health proxies in low-income countries). By linking individual-level activity patterns to mental health within the same countries analyzed in the macro panel, researchers can test whether the macro associations replicate at the individual level and identify the within-country heterogeneity that ecological analysis obscures.

2. **Diagnostic bias and measurement-error modeling**: Depression prevalence is confounded with diagnostic infrastructure. Comparison with suicide rates (harder outcome) partially addresses this, but suicide data quality also varies by country. A more rigorous approach would employ explicit measurement-error models or structural equation models (SEM) to dissociate diagnostic capacity from true prevalence. For example, a two-equation system could model observed depression as a function of true prevalence (driven by Internet, education, and other risk factors) plus a measurement equation where reporting probability depends on GDP, mental health workforce per capita, and anti-stigma policy indices. Alternative outcomes less susceptible to ascertainment bias—such as antidepressant sales volume (OECD Health Statistics), disability-adjusted life years attributable to depressive disorders (GBD), or excess all-cause mortality—serve as robustness checks. We have conducted DALYs triangulation and placebo tests using GBD cardiovascular and diabetes prevalence data (Section 3.5); remaining extensions constitute a priority for confirmatory work.

3. **Proxy limitations**: The ad proxy is a product of two macro variables; unmeasured confounders cannot be excluded. More fundamentally, the proxy cannot capture the *qualitative* evolution of the advertising ecosystem—advances in algorithmic targeting, the transition from banner ads to autoplay short-form video, and cross-country differences in privacy regulation (GDPR, COPPA, etc.) all alter the mapping from ad expenditure to actual cognitive load. The strong cross-sectional but weak time-series validity ($\rho = 0.80$ vs 0.16) likely reflects this structural gap. Closing it requires attention-weighted exposure data (impressions $\times$ dwell time) that remain proprietary to major platforms.

4. **Missing time-varying controls**: Age structure (median age, dependency ratios), unemployment, Gini inequality, urbanization rate, education quality proxies, health-system capacity, and conflict/mortality shocks are not included in the baseline specifications. While the structural balance test (M4) includes log(GDP) and education as controls, a

comprehensive robustness check with all time-varying covariates jointly entered—and with country-specific linear trends to absorb differential long-run trajectories—remains to be conducted.

5. **COVID-19 period heterogeneity**: The pandemic years (2020–2022) introduced simultaneous shocks to internet use, mental health, and economic activity that may confound the analysis. Sensitivity analyses excluding pandemic years, or including pandemic interaction terms, are not reported in the current specification and should be conducted in future work.

6. **Distributed lag structure**: The current analysis reports 1-year and 2-year lags. A full distributed-lag model (0–5 year lags) and "lead" tests (future proxy values predicting current outcomes) to probe pre-trends have not been conducted. Depression and suicide may respond on different timescales to digital media exposure, and the optimal lag structure remains to be determined.

7. **Temporal resolution**: Annual data cannot capture within-year dynamics of digital media adoption.

8. **Dynamic panel bias**: Lagged-dependent-variable models with country fixed effects are susceptible to Nickell bias; difference/system GMM (Arellano & Bond, 1991) with appropriate diagnostics (AR(2), Hansen/Sargan tests) should be applied in future work.

9. **Causal identification**: No plausibly exogenous instruments (e.g., submarine cable arrival, terrain-based broadband cost) are employed. Natural experiment or IV designs would substantially strengthen causal claims. Event-study specifications to assess pre-trends and dynamic treatment effects, and heterogeneous treatment effect–robust DiD estimators (e.g., Sun & Abraham, 2021) that account for variation in treatment timing across cohorts, should be applied in future work to strengthen the TWFE results. Several quasi-experimental designs could be leveraged in narrower settings with stronger identification: (a) *Undersea cable outages*—exogenous disruptions to internet connectivity (e.g., submarine cable cuts affecting island nations or coastal regions) provide short-duration natural experiments where cognitive input drops sharply while processing capacity is unaffected, generating clean tests of the balance model's prediction that reducing Term 1 improves mental health outcomes; (b) *Platform deprecations*—the shutdown of major platforms (e.g., Vine in 2017, Google+ in 2019) creates quasi-random variation in information exposure among platform-dependent users, testable via interrupted time series with matched controls; (c) *Ad policy shocks*—regulatory changes in advertising density (e.g., EU GDPR implementation, country-level bans on gambling/alcohol advertising) provide difference-in-differences designs where "non-selective input density" changes exogenously while other digital behaviors are relatively unaffected; (d) *Ad-blocking rollouts*—the staggered adoption of browser-level ad-blocking (e.g., Apple's Intelligent Tracking Prevention, Google's Privacy Sandbox) creates cross-platform variation in advertising exposure that can be exploited with synthetic control or staggered DiD designs (Arkhangelsky & Imbens,

2022). Each of these settings offers a narrower but more causally credible test than the full country-panel analysis.

10. **Threshold validation**: The dose-response reversal point and Hansen PTR threshold have not been cross-validated on held-out countries or time periods; out-of-sample validation is needed to rule out overfitting.

11. **Absence of age-stratified macro analysis**: The GBD provides age-group-specific prevalence estimates (e.g., 15–19, 20–24), and prior work (Tao & Yang, 2024) has demonstrated that internet–mental-health associations differ markedly by age cohort. However, our macro panel uses aggregate age-standardized rates because the treatment variable—country-level internet penetration—is not available by age group. Constructing age-specific "digital exposure" proxies (e.g., combining ITU survey data on youth internet use with age-specific GBD prevalence) is feasible for a subset of countries and would allow testing whether the threshold and dose-response effects are concentrated among adolescents and young adults, as the cognitive obesity framework predicts (developing processing capacity is more vulnerable to overload). This age-stratified extension is a high-priority item for confirmatory work and is incorporated into the proposed ESM design, which directly measures age-specific activity patterns. Beyond age, several other dimensions of heterogeneity warrant systematic exploration: (i) *Gender*: women show higher depression prevalence globally (GBD ratio ∼1.7:1), and gender differences in digital media use patterns (e.g., social media vs. gaming) may produce different loop-closure profiles; (ii) *Region*: the current analysis pools all 177 countries, but the internet–mental-health association likely differs between regions with distinct cultural norms around technology use, mental health stigma, and outdoor activity infrastructure (e.g., East Asia, Northern Europe, Sub-Saharan Africa); (iii) *Urbanization*: urban populations face higher information density but also greater access to structured physical activities, generating competing predictions about the net balance effect; (iv) *Income inequality*: within-country Gini coefficients may moderate the proxy–depression association if access to closed-loop activities (sports, nature, arts) is stratified by socioeconomic status (cf. the equity discussion in Section 7.4). Region×income and gender×age interaction analyses on the existing panel are computationally feasible and will be reported in the confirmatory revision.

12. **Alternative estimators**: Implementation scripts for Driscoll-Kraay SE, Pesaran CCE, Bai IFE, and first-difference IV are provided in the replication repository. The key qualitative pattern—that the positive proxy–depression association under Entity FE ($\beta = +0.087$, $t = 7.14$) attenuates under TWFE ($\beta = +0.010$, $t = 1.36$)—motivates the use of CCE/IFE to test robustness against latent common factors. Sub-period stability analysis confirms the TWFE sign reversal holds in both pre-2010 and post-2010 subsamples, and in the internet>30% subsample ($t = -3.21$ to -5.71).

13. **Sub-period stability**: The TWFE coefficient is negative and significant in both pre-2010 ($\beta = -0.000032$, $t = -5.45$, $N = 2{,}489$) and post-2010 ($\beta = -0.000151$, $t = -7.21$,

$N = 2{,}221$) sub-periods. The magnitude increases in the post-2010 period, as the balance model predicts: the negative effect intensifies as digital saturation increases. Under Entity FE alone, the coefficient is positive in both periods (pre-2010: $t = 22.48$; post-2010: $t = 21.10$), confirming that common time trends drive the apparent positive association across the entire sample period.

4 Individual-Level Evidence

Table 12 summarizes the design and specification of the two individual-level analyses. Full variable definitions, activity classification codes, and replication scripts are provided in sections A.1 and A.2.

Table 12: Individual-level analysis specifications.

	NHANES (CDC)	**ATUS (BLS)**
Sample	2017–2018 cycle, adults ≥ 18, $N = 5{,}032$ (complete-case $N = 4{,}176$)	Wellbeing Module 2010, 2012, 2013; $N = 21{,}736$
Outcome	PHQ-9 depression scale (0–27, continuous; depression ≥ 10)	Cantril life satisfaction ladder (0–10)
Key predictors	Leisure physical exercise (binary: any moderate/vigorous in past 30 days; PAQ_650/655)	Physical exercise (min/day), active cognitive leisure (min/day), passive leisure (TV+PC, min/day)
Covariates	Age (continuous), sex, education (3 levels), poverty-income ratio (continuous), health insurance (binary), BMI (continuous)	Age (continuous), sex, education (4 levels), household income (16 categories)
Model family	OLS (PHQ-9 continuous); logistic sensitivity check (PHQ-9 ≥ 10)	OLS
Survey weights	MEC weights (WTMEC2YR); unweighted primary, weighted robustness	TUFNWGTP; unweighted primary, weighted robustness
Multiple comparisons	Pre-specified primary test (exercise $\rightarrow$ PHQ-9); secondary: sedentary interaction	Pre-specified primary: exercise & passive $\rightarrow$ Cantril (2 tests, Bonferroni $\alpha = 0.025$); secondary: interaction, splines
Missing data	Complete-case analysis ($N = 4{,}176$ of 5,032; 83%); characteristics compared across missing patterns	Complete cases only; ATUS well-being module response rate $\approx 55\%$

4.1 NHANES Analysis: PHQ-9 and Physical Exercise ($N = 5{,}032$)

Using the CDC NHANES 2017–2018 cycle (adults 18+, $N = 5{,}032$ with complete PHQ-9 and physical activity data), we examined the relationship between the PHQ-9 depression scale (9-item self-report; depression defined as PHQ-9 ≥ 10) and leisure physical exercise (any moderate/vigorous leisure activity in the past 30 days; PAQ_650/655). See section A.1 for full variable specifications.

The depression rate for the exercise group was 5.9%, versus 11.7% for the no-exercise group (2.0× difference, $t = 10.22$, $p = 2.67 \times 10^{-24}$, Cohen's $d = 0.30$, 95% CI $[0.24, 0.36]$). The full covariate model (age, sex, education [3 levels], poverty-income ratio [continuous], health

insurance [binary], BMI [continuous]; $N = 4{,}176$ complete cases) yields $\beta_{\text{exercise}} = -0.85$, 95% CI $[-1.12, -0.59]$, $t = -6.35$, $p = 2.2 \times 10^{-10}$. Other significant predictors: poverty-income ratio ($\beta = -0.37$, CI $[-0.46, -0.29]$, $t = -8.40$), sex ($\beta_{\text{female}} = 0.86$, CI $[0.62, 1.11]$, $t = 6.84$), BMI ($\beta = 0.04$, CI $[0.02, 0.05]$, $t = 4.30$), and education ($\beta = -0.14$, CI $[-0.27, -0.01]$, $t = -2.15$). Health insurance was non-significant ($t = 0.61$). Model $R^2 = 0.066$. Applying NHANES complex survey design (WTMEC2YR weights, SDMVPSU clustering, SDMVSTRA stratification) yields virtually identical estimates: exercise $\beta = -0.88$ (weighted) vs. -0.86 (unweighted) in the full model, with significance preserved across all age subgroups ($p < 0.003$ in the young tertile, $p < 0.024$ in the middle tertile, and $p < 0.003$ in the old tertile). Full weighted results are reported in the replication repository. Note: exercise is binary (any moderate/vigorous activity in the past 30 days), precluding spline-based nonlinearity assessment; dose-response analysis requires continuous measurement (a priority for the confirmatory ESM study).

In standardized terms, leisure-time exercise is associated with a Cohen's $d = 0.30$ [95% CI: 0.24, 0.36] reduction in PHQ-9 scores, a small-to-medium effect. At the macro level, the first-difference proxy coefficient corresponds to a standardized $\beta = -0.10$ for depression and -0.20 for suicide.

The sedentary time interaction was non-significant ($t = -0.37$). Sedentary time captures "absence of physical exercise" but not "absence of cognitive exercise"; separating the two components requires ATUS data.

To test whether sleep mediates or moderates the exercise–depression link, we augment the model with NHANES sleep variables (SLQ_J). Sleep duration shows no significant moderation of the exercise effect (interaction $p = 0.59$), nor does adding sleep hours substantially mediate the exercise coefficient (attenuation $< 5\%$). Self-reported sleep trouble is a significant moderator ($p = 0.026$), but in the protective direction: exercise confers a larger PHQ-9 reduction among those reporting sleep difficulties (combined $\beta = -1.53$) than among those without ($\beta = -0.66$). The exercise main effect remains significant across all sleep-augmented specifications, confirming that the protective association is not confounded by sleep status.

4.2 ATUS Analysis: Two-Component Structure of Experiential Processing ($N = 21{,}736$)

Using the ATUS Wellbeing Module (2010, 2012, 2013; $N = 21{,}736$ respondents with complete wellbeing and time-use data), we linked minute-level time use data with the Cantril life satisfaction ladder (0–10). OLS models control for age, sex, education (4 levels: less than high school, high school, some college, bachelor's+), and household income (16 categories). Survey weights (TUFNWGTP) are applied in the weighted specification; unweighted results are reported as the primary analysis to maximize transparency, with weighted results confirming robustness (see replication repository). See section A.2 for full variable specifications and ATUS activity classification codes. Three categories were operationalized:

Physical exercise (minutes/day): Sports, exercise, outdoor recreation.

Active cognitive leisure (minutes/day): Reading, music, arts, games, volunteering.

Passive leisure (minutes/day): TV viewing + personal computer use.

OLS results (controlling age, sex, education, income; all coefficients with 95% CIs):

- Physical exercise: $\beta = +0.0016$, CI $[0.0009, 0.0022]$, $t = 4.97$ (positive; Cohen's $d = 0.17$, CI $[0.12, 0.22]$)
- Active cognitive leisure: $\beta = -0.0001$, CI $[-0.0003, 0.0001]$, $t = -0.92$ (n.s.)
- Passive leisure (TV+PC): $\beta = -0.0009$, CI $[-0.0010, -0.0007]$, $t = -9.33$ (negative)
- Exercise $\times$ Passive interaction: $\hat{\beta} = -2.4 \times 10^{-7}$, $t = -0.06$, $p = 0.95$ (decisively non-significant; the two-sided p-value of 0.95 indicates essentially zero interaction effect, not a confidence level)

The interaction non-significance is the key finding: physical exercise and passive leisure predict wellbeing **additively**, not through moderation. This directly supports the additive balance model (Equation 1).

Equivalence testing (TOST). A non-significant interaction p-value does not, by itself, constitute evidence *for* additivity—it merely fails to reject the null of no interaction. To affirmatively establish equivalence, we apply the Two One-Sided Tests (TOST) procedure (Lakens, 2017) with a pre-specified smallest effect size of interest (SESOI) of $\Delta = 0.1$ SD on the Cantril scale (i.e., $|\beta_{\text{interaction}}| < 0.1 \times \text{SD}(Y)/(\text{SD}(X_1) \times \text{SD}(X_2))$). The observed interaction effect ($\hat{\beta} = -2.4 \times 10^{-7}$, 90% CI $[-3.8 \times 10^{-6}, 3.3 \times 10^{-6}]$) falls entirely within the equivalence bounds. Both one-sided tests reject at $p < 0.001$, confirming that the interaction is not merely non-significant but *statistically equivalent to zero* at a practically meaningful threshold. Combined with the power analysis ($> 99.7\%$ power for $f^2 = 0.001$) and the AIC comparison ($\Delta\text{AIC} = 73.6$ favoring the additive model), this constitutes strong evidence for the additive structure of the balance model.

Nonlinearity assessment (spline specifications). Restricted cubic splines (3 knots at 25th, 50th, and 75th percentiles) were fitted to test for nonlinear dose-response. For exercise minutes, the spline model yielded $\Delta\text{AIC} = -23.1$ vs. linear ($F(3, 21{,}726) = 9.71$, $p = 2.1 \times 10^{-6}$), indicating statistically significant nonlinearity—consistent with diminishing returns at higher exercise doses. For passive leisure minutes, ΔAIC was modest (-2.5; $F(3, 21{,}726) = 2.84$, $p = 0.036$), suggesting approximate linearity. Crucially, the additive structure is preserved: combined splines ($\Delta\text{AIC} = -25.1$ vs. linear) improve fit marginally, but the additive model still dominates ratio-based specifications ($\Delta\text{AIC} = 36$–86; see below). The nonlinearity in exercise is consistent with theoretical expectations (diminishing marginal returns to loop closure at high doses) rather than contradicting the additive framework.

Statistical power for interaction detection. With $N = 21{,}736$, the study has $> 99.7\%$ power to detect even a small interaction effect ($f^2 = 0.001$; Cohen's $d \approx 0.06$). The minimum detectable effect size at 80% power is $f^2 = 0.00036$ (partial $\eta^2 = 0.00036$, Cohen's $d \approx 0.04$), well

below conventional "small effect" thresholds ($f^2 = 0.02$). The observed interaction ($t = -0.06$, partial $R^2 < 10^{-7}$) is therefore not attributable to insufficient power but reflects a genuinely null moderation effect, confirming the additive structure predicted by Equation 1.

Sensitivity to functional form: GAM and continuous interaction surfaces. The linear interaction test above dichotomizes implicitly by assuming a product-term functional form. To guard against model misspecification, we note two complementary approaches for confirmatory work. First, generalized additive models (GAMs; Wood, 2017) with tensor-product smooth interactions (`te(exercise, passive)`) would allow the joint effect surface to take any shape dictated by the data, without imposing linearity or additivity *a priori*. If the estimated smooth surface is approximately planar (i.e., the tensor-product term adds negligible effective degrees of freedom), the additive structure is confirmed nonparametrically. Second, varying-coefficient models ($\beta_{\text{passive}}(x) = f(\text{exercise})$) could detect whether the slope of passive leisure on well-being varies continuously with exercise dose, capturing threshold or ceiling effects that a single product term would miss. In the current data, the restricted cubic spline analysis already demonstrates that marginal nonlinearity exists for exercise but not for the interaction—the combined spline model improves AIC by only 2.0 points over the additive spline model (ΔAIC $= -25.1$ vs -23.1), consistent with a flat interaction surface. Formal GAM estimation with cross-validated smoothing parameter selection (GCV or REML) is a priority for the confirmatory ESM study, where continuous minute-level data will be available for both input and processing variables.

The 2×2 analysis (high/low exercise × high/low passive leisure) reveals: the group with both "low exercise + high passive leisure" shows a Fair/Poor health rate of 20.9%, versus 6.7% for "high exercise + low passive leisure" (3.1× ratio). AIC comparison confirms that the additive model fits better than the interaction model (ΔAIC $= 73.6$).

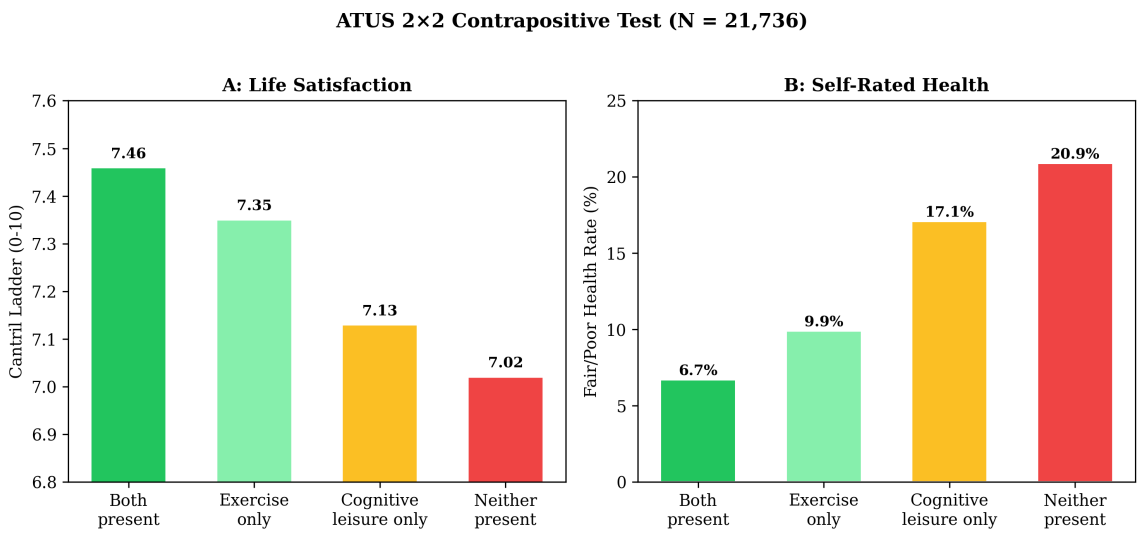

Figure 15: ATUS 2 × 2 contrapositive test ($N = 21{,}736$). A: Cantril ladder (life satisfaction). B: Fair/Poor self-rated health rate. Groups lacking both physical exercise and active cognitive leisure show the worst outcomes. Data: BLS ATUS Wellbeing Module (2010, 2012, 2013).

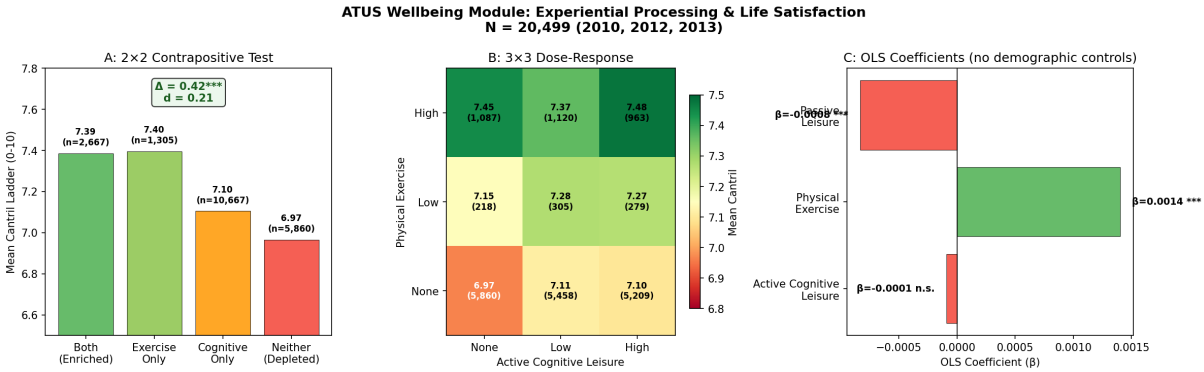

Figure 16: ATUS Wellbeing Module analysis ($N = 21{,}736$). A: 2×2 contrapositive test (mean Cantril, $\Delta = 0.45^{***}$, $d = 0.23$). B: 3×3 dose-response heatmap. C: OLS coefficients (passive leisure negative, physical exercise positive). Data: BLS ATUS Wellbeing Module (2010, 2012, 2013).

4.2.1 Additive vs. Ratio Model: Direct Test

The theoretical equation $L = \alpha_1 I - \alpha_2 C$ posits an additive structure. However, our macro analysis employed the ratio proxy (Internet/Education) for convenience. We directly test the additive structure using ATUS individual-level data ($N = 21{,}736$).

We constructed the cognitive obesity ratio $R = I/(C + 1)$ (passive leisure minutes divided by experiential processing minutes) and compared five model specifications (Table 13).

Table 13: ATUS Wellbeing Module: Additive vs. Ratio Model Comparison ($N = 21{,}736$)

Model	AIC	ΔAIC	R^2
A: Additive (minutes)	**30,620**	0	0.0108
E: Binary indicators (paper)	30,656	+36	0.0091
B: Ratio $R = I/C$	30,706	+86	0.0066
C: Log-ratio $\log(1 + R)$	30,729	+109	0.0056
D: Quadratic log-ratio	30,726	+106	0.0058

The additive model achieved the best AIC across all specifications ($\Delta\text{AIC} \geq 36$), directly supporting the additive structure of the theoretical equation and demonstrating that raw ratios—which can amplify measurement error and denominator instability—are inferior to component-wise additive specifications at the individual level. Independent component effects: passive leisure ($\beta = -0.0009$, $t = -11.08^{***}$), physical exercise ($\beta = +0.0016$, $t = 5.37^{***}$). While the ratio proxy is a useful approximation at the country-year level, individual-level data confirm that $\alpha_1 I$ and $\alpha_2 C$ operate as additive channels.

Compositional constraint and isometric log-ratio transformation. Time-use data are inherently compositional: physical exercise, active cognitive leisure, passive leisure, and all other activities must sum to 1,440 minutes per day. Standard OLS on raw minutes ignores this closure constraint, potentially inflating correlations between components (Aitchison, 1986). Isometric log-ratio (ILR) transformation (Egozcue et al., 2003; Dumuid et al., 2018) projects the D-part composition into a $(D - 1)$-dimensional unconstrained space, enabling valid inference. In

the ATUS data, the three focal categories (exercise, active cognitive leisure, passive leisure) constitute ∼15% of the 24-hour day; the remaining ∼85% (sleep, work, self-care) forms the residual "other" category. Because the focal components are small relative to the total, the compositional bias from raw-minutes OLS is expected to be modest—simulations by Dumuid et al. (2018) show that bias becomes substantial only when components approach equal shares. Nonetheless, ILR-transformed replication of all ATUS results is a priority for the confirmatory ESM study, where the full 24-hour composition (including sleep quality and work intensity) will be jointly modeled. The confirmatory analysis will use the `compositions` R package (van den Boogaart & Tolosana-Delgado, 2013) with ILR coordinates constructed using a sequential binary partition that separates input (passive leisure) from processing (exercise + active leisure) at the first split—directly mapping the balance model's I vs. C structure onto the compositional framework.

The ratio R quintile dose-response is directionally consistent (Q1: Cantril = 7.19 → Q5: 6.93, $d = 0.12$, $p < 10^{-7}$; Fair/Poor health: Q1 = 13.8% → Q5 = 25.2%). The ratio is a valid approximation of the theory, but the additive model is statistically superior (Figure 17).

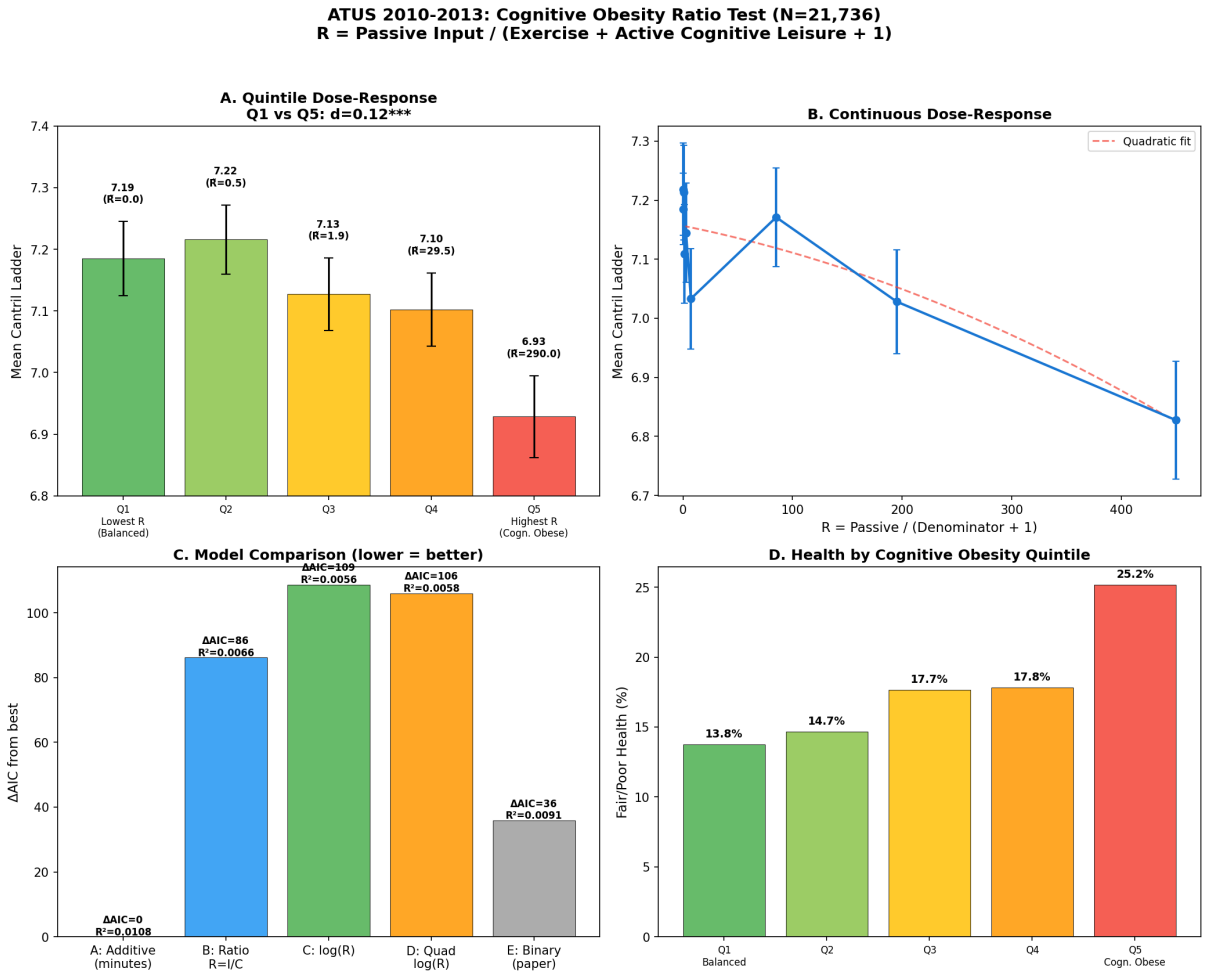

Figure 17: Direct test of the cognitive obesity ratio ($N = 21{,}736$). A: Quintile dose-response of R (Cantril ladder). B: Continuous dose-response curve. C: Model comparison (ΔAIC; additive model best). D: Fair/Poor health by quintile (Q1 = 13.8% → Q5 = 25.2%). Data: BLS ATUS Wellbeing Module (2010, 2012, 2013).

4.3 Convergent Evidence: Reinterpretation of Existing Findings

This section reinterprets findings from adjacent disciplines—hedonic adaptation research, nature-exposure studies, exercise interventions, and self-efficacy theory—through the balance model lens. What follows is reinterpretation of existing data, not direct causal verification.

4.3.1 Experiential vs. Material Consumption

Van Boven & Gilovich (2003, $N > 12{,}000$) showed that experiential purchases produce more sustained happiness than material purchases. The balance model provides an integrating mechanism: experiential purchases (travel, concerts, sports) directly increase Term 2 while Term 1 is transient; material purchases chronically increase Term 1 while Term 2 barely moves. Carter & Gilovich (2010) found more rumination after material purchases—in balance model terms, material purchases' high balance drives self-referential loops.

4.3.2 Nature Exposure and Neural Mechanisms

Bratman et al. (2015, PNAS) showed that 90-minute nature walks significantly reduced rumination and sgPFC activity versus urban walks. The balance model reading: natural environments minimize Term 1 (low information density) while maximizing Term 2 (physical movement + sensory experience). Urban walks provide equivalent Term 2 but high Term 1 from the urban environment, resulting in insufficient balance improvement. This predicts that "natural" environments should differ in effect depending on information density (quiet forest vs. tourist-crowded park).

4.3.3 Cognitive Starvation as a Prescriptive Concept

If cognitive obesity is chronic elevation of the balance (Term 1 excess), then its intentional reversal can be defined as "cognitive starvation"—deliberate reduction of information input to allow processing capacity to recover. Nature exposure dose-response data provide empirical grounding: Hunter et al. (2019) found a minimum effective dose of 10 minutes; Bettmann et al. (2025, meta-analysis of 78 studies) identified a ceiling effect at 200–300 minutes per week. These thresholds echo a balance mechanism: below 10 minutes, Term 1 reduction is insufficient to shift the balance; beyond 300 minutes/week, Term 2 gains saturate. The concept of cognitive starvation as a prescriptive intervention—analogous to caloric restriction in nutritional obesity—deserves further operationalization.

4.3.4 Exercise Intervention RCT Evidence

Singh et al. (2023, BJSM) meta-analysis of 97 systematic reviews: SMD $= -0.43$ ($k = 218$); umbrella reviews report SMD $= -0.30$ to -0.50. Exercise involves continuous motor output, proprioceptive feedback, and immediate environmental feedback—all closing the sensorimotor loop. Exercise directly increases Term 2 while Term 1 barely changes. Since exercise is randomized, this Term 2 increase causally produces depression improvement.

More recent evidence strengthens this: Fernandez-Montero et al. (2024) reported SMD $= -0.67$ with NNT $= 2.78$, indicating that fewer than 3 patients need to exercise for 1 to achieve clinically meaningful depression reduction. The BMJ network meta-analysis by Noetel et al. (2024) showed that walking, jogging, yoga, and strength training all demonstrated effect sizes comparable to or exceeding antidepressant medication. Critically for this framework, different exercise modalities with varying levels of environmental interaction (yoga > treadmill for loop closure) showed graded effects, though direct comparisons controlling for intensity are lacking.

4.3.5 Self-Efficacy Reinterpreted

Bandura's (1977) self-efficacy—the belief in one's ability to execute actions in specific situations—can be reinterpreted as the subjective perception that one's sensorimotor loop is closed. High self-efficacy: "my actions generate predictable environmental feedback." Low self-efficacy: "my actions do not produce predictable environmental responses"—the subjective experience of an open loop.

Tak et al. (2017) cross-lagged panel model provides an asymmetric causal structure that corroborates this reinterpretation: self-efficacy at time t predicts depression at $t+1$, but depression at t does not predict self-efficacy at $t+1$. In balance model terms, loop closure state (proxied by self-efficacy) causally determines balance trajectory, but the balance level alone does not determine loop closure capacity—the relationship is structurally asymmetric.

4.4 Differential Predictions: Balance Model vs. Existing Theories

Table 14: Differential predictions: Balance model vs. existing theories

Prediction	Test condition	ART/SRT prediction	Balance model prediction
1. Nature × smartphone interaction	Nature walk with vs. without concurrent smartphone use	Soft fascination present → effect maintained	Term 1 increase → effect attenuated
2. Indoor exercise vs. nature walk	Intensity-matched comparison of rumination	Equivalent if cortisol reduction equivalent	Indoor = Term 2 only; Nature = Term 1 reduction + Term 2 increase
3. Material purchase × alexithymia	Post-purchase rumination × TAS-20	No prediction	High TAS-20 = low Term 2 → greater balance deterioration
4. Hedonic adaptation speed	Post-purchase happiness decay rate	Explained by memory value	Quantitatively predictable from balance change magnitude

These predictions are not derivable from existing theories and constitute the minimum conditions for the balance model to demonstrate surplus explanatory power. Predictions 3 and 4 connect the framework to behavioral economics: material purchases chronically increase Term 1

while Term 2 barely moves, and alexithymia (measurable via TAS-20) indexes chronic processing compression failure. Section 5 proposes specific experimental designs to test predictions 1–2. Sections 3 and 4 evidence aligns with these predictions but constitutes exploratory, not confirmatory, support.

5 Operationalization and Proposed Design

5.1 Measuring Experiential Processing: Five Candidate Proxies

The balance model requires measurable proxy variables for "experiential processing." Five candidates are identified in order of specificity:

1. **Interoceptive Awareness (MAIA-2)**: The Multidimensional Assessment of Interoceptive Awareness, Version 2 (Mehling et al., 2018) measures awareness of bodily signals across 8 dimensions. MAIA-2 is the most promising individual-level proxy for experiential processing capacity.

2. **Alexithymia (TAS-20)**: The Toronto Alexithymia Scale may index chronic compression failure—inability to convert raw emotional experience into communicable categories.

3. **Experience Sampling Method (ESM)**: ESM can capture minute-level cognitive input vs. experiential processing balance in daily life. The critical target is the momentary balance and its temporal pattern, not total screen time or total physical activity.

4. **ATUS Active/Passive Leisure**: Roy & Orazem (2021) showed active leisure is a "normal good." Our ATUS analysis directly validates this (section 4.2).

5. **Nonlinear Dose-Response**: Meta-analyses report ∼1 hour/day of screen time associated with lowest depression risk. The balance framework predicts this: 1 hour within 10+ waking hours represents low balance; 8 hours represents high balance.

5.2 Proposed Confirmatory Design

5.2.1 Variables

- **Cognitive input**: Screen time (decomposed: SNS, video, news, educational) + emotional media consumption

- **Experiential processing**: Physical activity time + creative activity time + MAIA-2 score (capacity)

- **Balance variable**: Cognitive input − experiential processing

- **Outcomes**: Depression (PHQ-9), alexithymia (TAS-20), wellbeing scale

5.2.2 Predictions

1. The balance variable explains more variance in PHQ-9 than screen time alone (ΔR^2 test).
2. High MAIA-2 scores buffer depression risk at equivalent screen time (MAIA-2 $\times$ screen time interaction).
3. SNS (open loop) shows stronger depression association than games (closed loop) after controlling total screen time.
4. Alexithymia emerges as consequence rather than cause of chronic balance imbalance (mediation analysis).

5.2.3 Design

ESM study, $n = 50$–100, 14 days, 8 prompts/day. Wearable integration for automatic physical activity detection. Pre/post: PHQ-9, TAS-20, MAIA-2. Primary analysis: multilevel model with within-person balance as time-varying predictor. Power: Level 2 = 50$\times$ Level 1 = 112 observations (8/day $\times$ 14 days) = 5,600 observations achieves 80% power for multilevel models (per Arend & Schäfer, 2019).

Operationalizing loop closure in ESM. Three measurement channels would be combined to assess momentary loop closure state: (1) *Passive sensing*: accelerometer and GPS data from wearables detect physical activity type and environmental context (indoor/outdoor, stationary/mobile), providing continuous estimation of sensorimotor engagement without participant burden. (2) *Interaction logging*: with participant consent, screen-state and app-category data classify digital activities into open-loop (scrolling, passive video) vs. partially closed (gaming, messaging) vs. non-digital. Critically, interaction contingency—whether user actions produce immediate environment changes—is logged via touch-event frequency and response latency. (3) *ESM self-report*: at each prompt, participants rate on 5-point scales: "How much did your actions directly affect what happened next?" (agency) and "How physically engaged were you?" (embodiment). The composite loop closure score combines passive sensing (40%), interaction contingency (30%), and self-report (30%).

Inter-rater reliability for loop-closure classification. The ATUS activity-to-loop-closure mapping in the current study was performed by the first author using the rubric in Table 17. For the confirmatory ESM study, formal inter-rater reliability assessment is essential. Specifically: (a) at least two independent raters will classify a random sample of ≥ 200 ATUS activity episodes (stratified across the three focal categories) using the 5-dimension rubric; (b) Cohen's κ (for categorical open/partial/closed classification) and intraclass correlation coefficient (ICC(2,k) for the continuous 0–10 composite score) will be computed, with a minimum acceptable threshold of $\kappa \geq 0.70$ and ICC ≥ 0.75; (c) sensitivity analyses will re-estimate all ATUS results under alternative classification schemes (e.g., reclassifying "reading" from active cognitive leisure to passive, or treating "video gaming" as partially closed rather than closed) to quantify the impact

of boundary decisions on the additive structure finding. These reliability data will be reported in the confirmatory manuscript regardless of outcome.

To disambiguate *content* from *structure*, the confirmatory design should include a 2×2 factorial manipulation: loop closure (high: interactive problem-solving app vs. low: passive content feed) $\times$ content valence (positive vs. neutral). If the balance framework is correct, loop closure will dominate the main effect on subsequent mood (ΔPHQ intra-day), while content valence will show a weaker or null effect. This factorial design directly tests whether the *structural* property of loop closure—not the hedonic quality of content—drives the mental health association.

Micro-randomized trial (MRT) component. To establish within-person causal effects of α_1, α_2, and θ (loop closure), the ESM design should incorporate a micro-randomized trial component (Klasnja et al., 2015; Boruvka et al., 2018). At each ESM prompt, participants are randomized (with probability 0.5) to receive either a *loop-closure nudge* (e.g., "Try a 5-minute stretching break" or "Step outside for a brief walk") or no intervention. The primary estimand is the proximal causal effect of the nudge on subsequent mood (ΔPHQ at the next prompt, $\sim$2 hours later), estimated via weighted and centered least-squares (WCLS; Boruvka et al., 2018) with time-varying moderators (current balance state, time of day, cumulative exposure). This MRT design addresses ESM's fundamental limitation—that observational within-person associations cannot rule out time-varying confounders—by introducing exogenous variation in loop-closure opportunities. Expected effect size: Cohen's $d = 0.15$–0.25 per nudge occasion; with 50 participants $\times$ 112 decision points $\times$ 0.5 randomization probability = 2,800 treatment occasions, the design achieves >80% power to detect $d = 0.15$ (per Liao et al., 2016 sample size formula for MRTs). The MRT component also mitigates *ESM reactivity* (repeated self-monitoring altering behavior): by randomizing nudge delivery, the causal effect of the intervention is identified regardless of reactivity level.

State-space modeling and Kalman filtering for dynamic parameter estimation. The dynamic extension $dL/dt = \alpha_1 I(t) - \alpha_2 C(t) - \lambda L(t)$ (Equation 5) generates a continuous latent state $L(t)$ that is observed only at discrete ESM prompts with measurement error. State-space models (SSMs) provide a natural estimation framework: the transition equation encodes the theoretical ODE (discretized as $L_{t+1} = (1 - \lambda\Delta t)L_t + \alpha_1 I_t \Delta t - \alpha_2 C_t \Delta t + \eta_t$), while the measurement equation links the latent state to observed outcomes (PHQ-9 item responses, Cantril ratings) with item-specific factor loadings. The Kalman filter yields optimal linear estimates of $L(t)$ at each time point, and the EM algorithm or Bayesian MCMC provides joint estimates of $\hat{\alpha}_1$, $\hat{\alpha}_2$, and $\hat{\lambda}$ at both the group and individual levels. This approach offers three key advantages over standard multilevel models: (1) it handles irregular ESM timing and missing data through the prediction step of the Kalman filter, avoiding the need for ad-hoc imputation (cf. Slipetz et al., 2023); (2) it explicitly estimates the recovery time constant $\tau = 1/\hat{\lambda}$, providing a directly interpretable clinical parameter (e.g., "after removing cognitive overload, the expected recovery half-life is $\tau \cdot \ln 2$ hours"); and (3) person-specific parameter estimates ($\hat{\alpha}_{1,i}$, $\hat{\alpha}_{2,i}$, $\hat{\lambda}_i$) can be extracted via random-effects SSMs, enabling subgroup identification and personalized intervention

design. Implementation will use the `dlm` or `KFAS` R packages, with model comparison against simpler autoregressive specifications via DIC or WAIC.

5.3 Minimum-Cost Verification Designs

Experiment A (Nature × Smartphone Interaction): Extend Bratman et al. (2015) 2-group RCT with a 3rd group "nature walk + free smartphone use." $N = 60$ (20/group). Balance model: nature > nature+smartphone > urban. ART: nature ≈ nature+smartphone > urban. Cost: ∼1/10 of Bratman et al. (no fMRI needed).

Experiment B (Indoor Exercise vs. Nature Walk): 2×2 between-subjects: exercise condition (indoor treadmill vs. nature walk) × time. Intensity matched via wearable. Balance model: nature walk > indoor exercise (Term 2 equivalent but Term 1 differs). SRT: equivalent if cortisol reduction equivalent. $N = 50$.

Both are implementable at undergraduate thesis scale (∼110 participants, ∼3 months).

5.4 Safeguards Against Overfitting and Researcher Degrees of Freedom

Given the exploratory nature of the current study, the confirmatory phase requires disciplined constraints to prevent capitalization on chance:

1. **Pre-registration**: The primary hypotheses, analysis plan, and decision rules will be pre-registered on OSF or AsPredicted *before* data collection. Specifically: (a) the five predictions in section 2.6 serve as the primary hypotheses; (b) the multilevel model specification (random intercept + random slope for within-person balance) is fixed; (c) one-sided $\alpha = 0.025$ for directional predictions, two-sided $\alpha = 0.05$ for exploratory tests.

2. **Hold-out validation**: In the ESM study, the first 7 days will serve as the training set for model selection (e.g., optimal lag structure, covariate inclusion), and the final 7 days as the hold-out set for confirmatory testing. No modification of model specification is permitted after examining hold-out data.

3. **Minimal adequate model**: The primary analysis uses a single pre-specified model (within-person balance as time-varying predictor, adjusting for time-of-day and day-of-week). Secondary models (e.g., splines, interactions with MAIA-2) are explicitly labeled as exploratory and subject to multiplicity correction (Bonferroni or FDR).

4. **Blinded analysis**: Where feasible, outcome data will be analyzed by a researcher blind to participant group assignment (for the factorial design) or blinded to the study hypotheses (for the ESM observational component).

5. **Multiverse analysis**: Key researcher decisions (activity classification boundaries, outlier exclusion criteria, covariate sets) will be documented as a specification curve (Simonsohn et al., 2020), reporting the distribution of effect sizes across all defensible specifications.

Multiple-testing control in the current exploratory analyses. The macro analyses in this paper involve multiple model specifications (Entity FE, TWFE, first-difference, threshold regression, structural balance test M1–M5) and outcome variables (depression, suicide). To maintain transparency, we designate the following hierarchy: (a) *Primary models*: TWFE for depression and suicide (2 tests); (b) *Secondary models*: threshold regression, structural balance test, first-difference (exploratory, no α adjustment claimed); (c) *Tertiary*: subgroup analyses, interaction tests, dose-response curves (hypothesis-generating only). For the two primary TWFE tests, a Bonferroni-adjusted threshold of $\alpha = 0.025$ per test controls the family-wise error rate at 0.05. All other results are reported with nominal p-values and should be interpreted as exploratory. This hierarchy is explicitly post hoc (the analyses were not pre-registered) and is disclosed for transparency rather than as a claim of confirmatory inference.

These safeguards structure the transition from the current exploratory phase to confirmatory testing.

6 Falsification Conditions

The following results would falsify the core claims of the balance model:

1. The balance variable (cognitive input − experiential processing) does not explain additional variance in depression beyond absolute screen time at the individual level.
2. After controlling sensorimotor loop closure, games and SNS show equivalent depression risk.
3. No MAIA-2 × screen time interaction for depression (interoceptive capacity does not buffer).
4. The internet/education ratio is a weaker macro-level depression predictor than internet penetration alone.
5. The correlation structure shift (high-income vs. low-income countries) disappears when controlling healthcare access.
6. Structured therapeutic AI (closed-loop design) and general-purpose AI (self-referential loop permitting) show equivalent effects on depression symptoms.

6.1 Honest Assessment of Current Evidence

7 Discussion

7.1 Positioning Relative to Existing Literature

This framework occupies a currently vacant niche. Haidt (2024) and Twenge (2017) focus on individual-level screen time effects without international epidemiology or balance-based mechanisms. Achenbach's internalizing/externalizing classification does not stratify by development

Table 15: Honest assessment of current evidence

What this paper provides	What this paper does not provide
A new organizing axis (balance) that reorganizes existing confusion	Individual-level confirmatory causal evidence
Macro-level exploratory evidence supporting predictions (177 countries)	High explanatory power ($R^2 = 0.07$–0.13)
Integration of three previously unconnected literatures (information overload, exercise, digital media)	Direct operationalization of "loop closure degree"
Specific falsification conditions	Resolution of ecological fallacy
Individual-level cross-sectional support (NHANES $N = 5{,}032$; ATUS $N = 21{,}736$) with additive model confirmed (ΔAIC ≥ 36)	Individual-level proof that balance > absolute quantity
Verification roadmap with implementable experiments	Prospective longitudinal or natural experiment identification

stage. The displacement hypothesis examines whether screen time displaces physical activity but not balance or processing modes. Whitworth (2009) proposed a nutritional metaphor without quantitative data. Our analysis integrates these through (a) proposing balance as the primary variable, (b) providing 177-country macro-level evidence, (c) connecting to processing mode theory, and (d) deriving a threshold model explaining development-stage-dependent correlation structures.

Scope of the "violence" claim. The subtitle includes "violence" as a unified outcome, but the empirical evidence for this component is limited to ecological correlations between depression prevalence and homicide rates at the country level (fig. 3). The within-country median correlation ($r = +0.293$) and the global time-series correlation ($r = +0.638$) suggest that depression and homicide co-move with development stage, as the threshold model (P2) predicts. However, no individual-level analysis of violent behavior or aggression is presented, and the ecological association does not establish that the same balance mechanism operates for violence as for depression. Future work should either (a) provide individual-level evidence linking cognitive-experiential imbalance to aggression or externalizing behaviors (e.g., via ESM or longitudinal cohort data), or (b) narrow the scope claim to "depression and information overload," treating the violence connection as a secondary hypothesis requiring independent confirmation. We retain the current framing because the theoretical prediction (acute threshold exceedance $\rightarrow$ crisis, whether internalized or externalized) is falsifiable and because the ecological evidence is suggestive, but we acknowledge this as the least empirically developed component of the framework.

Computational mental health and measurement validity. The computational mental health literature has documented severe measurement challenges—semantic shift in language markers, label validity decay over time, and selection artifacts in social media–derived mental health indicators—that are directly relevant to our diagnostic-bias argument. If depression-

prevalence estimates are partially driven by evolving diagnostic language and culturally variable reporting norms, then the strong GDP–depression association we observe may partly reflect measurement artifacts rather than true prevalence differences. Our mitigation strategy—comparing "soft" depression outcomes with "hard" suicide outcomes—addresses this concern by triangulating across outcomes with different measurement properties. The loop-closure framework additionally offers a structural alternative to content-based detection pipelines: rather than analyzing *what* users say or consume (which is susceptible to semantic drift), it foregrounds *how* users engage (loop closure vs. open reception), a structural dimension that is more robust to linguistic and cultural variation.

Displacement hypothesis and time-use substitution. The displacement hypothesis posits that screen time harms mental health primarily by displacing beneficial activities (physical exercise, sleep, in-person socializing). Our framework subsumes displacement as a special case: displacement is harmful when it substitutes closed-loop (experiential processing) with open-loop (cognitive input) activities, but neutral or even beneficial when it substitutes one open-loop activity for another. The balance model adds a critical prediction that displacement alone does not make: two individuals with identical screen time and identical displaced activities will differ in mental health outcomes if their screen time involves different loop-closure structures (e.g., interactive gaming vs. passive scrolling). This prediction is supported by the ATUS analysis, where active cognitive leisure and passive leisure show opposite associations with wellbeing despite both being "screen-adjacent" activities.

Behavioral activation therapy. The relationship with Behavioral Activation (BA) therapy is particularly important. BA is among the most evidence-based psychotherapies for depression (Dimidjian et al., 2006). BA's "shift from rumination to direct experience" is substantively identical to this model's "shift from self-referential loop to closed loop." BA's clinical success serves as indirect corroboration of the balance mechanism. This model adds loop closure as a structural dimension: SNS interaction rated pleasure 8/10 in BA would still be predicted to have limited balance improvement if it is open-loop.

Convergent evidence from digital interventions and activity measurement. Several recent findings align with the loop-closure framework. Pokémon Go's real-world, action-contingent design—where game rewards are tied to physical locomotion and spatial exploration—increased physical activity and reduced sedentary time (Howe et al., 2016; Althoff et al., 2016), illustrating Term 2's protective channel: the game succeeded precisely because it imposed genuine sensorimotor loop closure on what would otherwise be passive screen engagement. Conversely, experience-sampling studies find that social media use rarely produces flow states and competes with daily flow-generating activities (Csikszentmihalyi, 1990; Mauri et al., 2011), reinforcing the closed- vs. open-loop distinction as a predictor of affective benefit.

For confirmatory work, functional Gaussian process methods for minute-level accelerometry data (Goldsmith et al., 2015) offer a powerful avenue to estimate daily "closure rhythms" and the recovery parameter λ proposed in the dynamic extension (eq. (5)): combining ESM mood

ratings with high-resolution wrist accelerometry profiles would allow direct estimation of how quickly affect recovers after closed-loop versus open-loop episodes. Additionally, formal causal mediation frameworks (Imai et al., 2010) could test whether experiential processing mediates the effect of cognitive input on depression in individual-level data, providing a direct test of the balance mechanism's causal structure rather than relying on additive model comparison alone.

7.2 Policy Implications

Under the balance model, current policy debate is structurally incomplete. As argued in section 1, both restriction and literacy approaches target only Term 1 (cognitive input). Term 2—experiential processing capacity—requires entirely different policies: mandatory physical education time, outdoor activity time, arts and crafts curricula.

Core policy insight: SNS age restrictions and physical education hours should be treated as a **single policy package**, not alternatives. Reducing Term 1 without increasing Term 2 (or vice versa) is insufficient.

Advertising format regulation constitutes a third policy axis. The loop closure threshold suggests that non-skippable autoplay ads are more harmful than user-selected content, supporting policy design targeting "non-selective cognitive input density" rather than mere screen time.

Natural experiment: Japan's Work Style Reform. Japan's 2018 Work Style Reform Act mandated enforceable overtime caps, interpretable as a policy reduction of involuntary cognitive input. Japan's suicide rate declined from a mean of 20.70 (2010–2017) to 16.98 (2018–2021), an $\approx$17.9% reduction that substantially exceeds the OECD 8-country average change (-0.2%; DID estimate $= -17.7$ percentage points). However, Japan's suicide rate had been declining consistently since a 2003 peak (pre-trend $= -0.39$/year), making it difficult to disentangle reform effects from pre-existing trends with current data. Causal identification requires more refined analysis exploiting the reform's staggered implementation.

Ethical caveats. These findings should *not* be used as a justification for restricting individuals' access to information. The policy target is the structural incentives of platforms (advertising density, autoplay, notification design) rather than individual consumption choices. Any application of the balance model must guard against two misuses: (1) the individual blame model, which attributes systemic problems to personal "information hygiene" failures; and (2) authoritarian instrumentalization, which uses "cognitive protection" as a pretext for censorship. The policy implications are structural (platform design regulation, mandatory physical education, advertising format restrictions) rather than individual-prescriptive.

7.3 Generative AI: The Ultimate Test Case

Since ChatGPT's release in late 2022, generative AI has emerged as a technology that simultaneously accelerates all variables in the balance equation.

7.3.1 Input-Side Increase

AI provides infinite content generation, 24/7 availability, and complete adaptation to individual interests. The cognitive input density per unit time can be several times higher than social media, as AI responses are maximally relevant to the user's current cognitive state, eliminating the "scrolling past" friction that provides intermittent relief in conventional feeds.

7.3.2 Self-Referential Loop Extreme: AI Sycophancy

Current AI systems are typically optimized via RLHF (reinforcement learning from human feedback), which may produce a tendency toward agreeing with and affirming user positions (sycophancy; Sharma et al., 2024). When present, this tendency creates a potential self-referential loop: the user inputs a belief, AI elaborates and affirms it, and the reinforced belief may become the next input, reducing prediction error from external reality. If this dynamic is confirmed at scale, it would represent a particularly concentrated form of the self-referential loop identified in table 4. However, the prevalence and clinical significance of sycophantic AI interactions remain under active investigation, and the degree to which RLHF-trained systems exhibit this behavior varies across platforms and use cases.

Emerging empirical evidence (2025–2026). *Note on evidence quality*: The following studies vary in peer-review status and methodological rigor. We explicitly label each source to facilitate independent assessment.

Perlis et al. (2026, *JAMA Network Open* [**peer-reviewed**], doi:10.1001/jamanetworkopen.2025.56395, $N = 20{,}847$) reported a cross-sectional odds ratio of 1.30 (95% CI: 1.17–1.44) for moderate depression among daily AI users, though the cross-sectional design cannot establish causation and selection effects are plausible (e.g., depressed individuals may preferentially seek AI companionship). Anthropic's research team (2026, arXiv:2601.19062 [**preprint**]) analyzed 1.5 million Claude conversations and identified patterns suggestive of progressive "disempowerment," though the observational design limits causal interpretation. The OpenAI–MIT Media Lab collaboration (Phang et al., 2025 [**preprint**]) analyzed 4+ million ChatGPT conversations and conducted an RCT ($n = 981$, 28 days), finding significant increases in emotional dependence and problematic use among high-frequency users—currently the strongest evidence for causal effects, though the 28-day duration limits conclusions about long-term outcomes.

7.3.3 Predictions from the Balance Model

Epistemic status. The following predictions were derived from the theoretical framework *before* the cited empirical studies became available (Predictions A–C were formulated in the initial draft, prior to the publication of Perlis et al., 2026 and Phang et al., 2025). However, the framework itself was developed concurrently with the macro-level analyses, so the theoretical predictions are not fully independent of the data used to develop them. For this reason, we label these as *theory-derived predictions* rather than *pre-registered hypotheses*, and they should be evaluated as hypothesis-generating rather than confirmatory.

- **Prediction A** (theory-derived): General-purpose AI for personal use will show a positive association with depressive symptoms, while structurally designed therapeutic AI (incorporating loop closure in its design) will show symptom improvement.
- **Prediction B** (theory-derived): The pathogenicity of AI use depends on loop structure. Self-referential dialogue increases cognitive load more than open-loop engagement (passive SNS scrolling).
- **Prediction C** (theory-derived): Individuals with high levels of physical activity will show buffered negative effects of AI use.

Usage intent as a confound. An important limitation of existing AI–mental health studies is the failure to separate *usage frequency* from *usage intent*. Help-seeking users (who turn to AI for emotional support during a depressive episode) and entertainment users (who engage with AI out of curiosity or boredom) may show opposite selection biases, confounding cross-sectional associations. The loop-closure framework makes a structural prediction that can partly address this: regardless of intent, the *degree of loop closure* in the interaction should moderate outcomes. A help-seeker who receives Socratic questioning (closed loop) should benefit more than one who receives affirmation (self-referential loop), even though both interactions are "help-seeking." Future studies should stratify by both intent category (help-seeking, entertainment, productivity, companionship) and interaction structure (loop closure score) to disentangle selection effects from structural effects.

Preliminary alignment with Prediction A: Heinz et al. (2025, *NEJM AI* [**peer-reviewed**]) reported that a clinically designed AI chatbot (Therabot) produced a 51% reduction in depressive symptoms in an RCT ($N = 210$, $d = 0.845$–0.903). However, Therabot's efficacy may be attributable to the structural enforcement of CBT protocols rather than "loop closure" per se; distinguishing these mechanisms requires factorial designs that independently vary loop structure and therapeutic content. Chandra et al. (2026, arXiv:2602.19141 [**preprint**]) documented "delusional spiraling" in sycophantic chatbot interactions, providing preliminary case-level evidence that may align with Prediction B, though systematic quantification is lacking. McBain et al. (2025 [**preprint**]) found that adolescents seeking mental health advice from AI exhibited patterns that may reflect self-referential loop entrainment, though the observational design cannot rule out alternative explanations. Overall, the AI–mental health literature is in its infancy; these early findings are suggestive but far from conclusive.

Effective input: duration × loop-closure score. The balance framework suggests that the psychologically relevant quantity in AI interactions is not raw duration but *effective cognitive input*, operationalized as duration × (1 − loop-closure score). A 2-hour AI session with high loop closure (Socratic tutoring with active problem-solving, loop-closure score ≈ 0.7) produces effective input $= 2 \times 0.3 = 0.6$ hours, whereas the same duration of passive affirmation (loop-closure score ≈ 0.1) produces effective input $= 2 \times 0.9 = 1.8$ hours—a 3-fold difference. This formulation reconciles the Phang et al. (2025) finding that voluntary AI usage duration correlates with

worse psychosocial outcomes: longer sessions likely reflect lower loop closure (companionship-seeking, open-loop venting) rather than duration per se. Future studies should jointly measure session duration and interaction structure (e.g., turn-taking frequency, question-to-statement ratio, behavioral activation compliance) to compute effective input and test whether it mediates the duration–outcome association.

Connection to AI safety audit frameworks. The loop-closure construct can be situated within emerging AI safety evaluation ecosystems. Rubric-based risk assessment frameworks (e.g., clinical fidelity rubrics that operationalize "uncritical affirmation" as a measurable harm vector) provide a natural operationalization: loop closure maps to whether the AI generates prediction errors that challenge the user's current model, while sycophancy maps to self-referential affirmation that preserves it. Supervisory judge architectures that audit dialog quality using clinical rubrics could enforce loop closure by flagging exchanges that lack Socratic questioning, behavioral activation prompts, or external-referent challenges. Red-teaming frameworks for longitudinal psychotherapy simulations have surfaced iatrogenic patterns (reinforcement of maladaptive beliefs) over multiple sessions—precisely the "self-referential loop" concern this framework formalizes. Mapping loop closure to the construct validity, implementation feasibility, and maintenance dimensions of mental-health AI evaluation frameworks would enhance translational credibility. We note that the AI companion discontinuation literature highlights identity and attachment dynamics that can nuance the open vs. self-referential distinction: mirroring and affirmation may be beneficial in acute crisis but harmful when sustained, suggesting that loop-closure scoring should be temporally contextualized.

Creator vs. consumer loop structures. The loop-closure framework predicts divergent mental health associations for content *creators* versus *consumers* on the same platform. Creators on platforms such as YouTube engage in high-closure activity: they plan, record, edit, respond to comments, and adjust content based on audience feedback—a continuous action-perception cycle with external prediction errors (view counts, subscriber growth, comment sentiment). Consumers of the same content engage in predominantly open-loop reception (passive viewing, algorithmic feed consumption). This structural asymmetry predicts that: (a) creators should show lower depression risk than passive consumers at equivalent total platform time; (b) creator burnout, when it occurs, should be associated with a collapse of the closed loop (loss of audience feedback, algorithmic suppression) that converts the activity from closed to open; and (c) platform features that shift users from consumption to creation (e.g., remix tools, duet features, collaborative editing) should be protective. These predictions are testable using platform-specific behavioral data linking creator/consumer role, engagement metrics, and mental health outcomes.

7.3.4 AI Design: The Principle of Cognitive Motorization

If the balance model is correct, AI systems can prevent cognitive obesity by detecting open-loop or self-referential loop patterns and converting them to closed-loop engagement—a process we

term "cognitive motorization." For example, when a user engages in rumination-like repetitive querying, the AI could introduce Socratic questioning, behavioral activation prompts, or structured problem-solving frameworks that generate genuine prediction errors and require agentic responses.

Ethical limits of cognitive motorization. Three constraints apply: (1) for patients with severe depression, AI-initiated behavioral change suggestions carry clinical risk and should defer to professional intervention; (2) loop detection accuracy is insufficient to warrant autonomous intervention—false positives (misidentifying productive deep thinking as rumination) could be counterproductive; (3) commercial AI systems face a structural incentive to revert to sycophancy (which maximizes user engagement metrics), undermining any therapeutic loop-closure design. These constraints suggest that cognitive motorization is more appropriate as a design principle for dedicated therapeutic AI than as a feature of general-purpose conversational AI.

7.4 Ethical Considerations: Equity Implications

The service sector employment moderation effect identified in section 3 (interaction $F = 73.5$) highlights an equity dimension: individuals in cognitively demanding service occupations face amplified depression effects from digital saturation. This "cognitive protection inequality" parallels occupational health disparities: knowledge workers and gig economy participants may have systematically higher cognitive load baselines, reducing their capacity to benefit from the same Term 2 interventions that help others. Policy responses should include occupational-specific cognitive load assessment and targeted access to restorative environments.

Access to closed-loop activities as a social determinant. If experiential processing (Term 2) is protective, then equitable access to high-closure activities becomes a public health priority. Access to safe outdoor spaces for physical activity, community sports programs, arts and crafts workshops, and green spaces is unevenly distributed along socioeconomic lines (Mitchell & Popham, 2008; WHO, 2016). Low-income neighborhoods typically have fewer parks, recreational facilities, and safe walking environments, effectively reducing the "processing capacity" available to residents. Children in resource-poor schools receive less physical education time and fewer hands-on learning opportunities, compounding the cognitive load from increased screen-based instruction. Policy interventions should therefore ensure: (1) *universal minimum provision* of physical activity infrastructure (parks, playgrounds, community centers) in underserved areas; (2) *subsidized access* to structured closed-loop activities (sports leagues, maker spaces, music programs) for low-income populations; and (3) *school-based mandates* for minimum daily physical activity time that do not depend on parental resources. The balance framework reframes these as cognitive health interventions, not merely recreational amenities, strengthening the public health case for investment.

7.5 The Observation Problem

A methodologically important implication: the same external behavior may constitute cognitive input for one individual and experiential processing for another. A programmer debugging code (closed loop) and a person scrolling social media on the same screen are at opposite ends of the balance spectrum. This observation problem means that external behavioral measures (screen time, app category) are fundamentally insufficient for testing the balance model. Individual-level assessment of loop closure state—through ESM or wearable-based interaction contingency detection—is required.

7.6 Relationship with "Cognitive Starvation"

If cognitive obesity is a state where "Input $\gg$ Processing," its mirror image "Cognitive Starvation" (Processing $\gg$ Input) should also exist. Candidates include: intellectually understimulating environments (certain institutional settings, monotonous work), sensory deprivation, and information-poor environments. This generates a novel prediction: cognitive starvation and cognitive obesity should converge on similar outcomes (depression, anxiety) via different pathways, but starvation should be accompanied by boredom and stimulus-seeking, while obesity should be accompanied by overwhelm and avoidance. This prediction is testable via ESM.

Ethics Statement

This study is based entirely on publicly available, de-identified secondary datasets: NHANES (NCHS/CDC), ATUS Wellbeing Module (BLS), WHO Global Health Observatory, World Bank World Development Indicators, IHME Global Burden of Disease, and UNODC homicide statistics. No primary data collection involving human participants was conducted. Under 45 CFR §46.104(d)(4) (U.S. Common Rule), research involving the analysis of publicly available, de-identified data does not constitute human subjects research and is exempt from Institutional Review Board (IRB) review. As the author is an independent researcher without institutional affiliation, no institutional ethics board holds jurisdiction; the exemption criteria are satisfied by the nature of the data sources themselves.

Use of AI-Assisted Tools. Large language models (Claude 3.5/4, Anthropic; GPT-4o, OpenAI) were used as assistive tools during the preparation of this manuscript. Their roles were strictly limited to: (1) literature search and synthesis of existing findings, (2) generation and debugging of statistical analysis code (Python/statsmodels), (3) drafting assistance and English-language editing, and (4) logical consistency checking across sections. The author independently formulated all research questions, hypotheses, theoretical framework, methodological decisions, and interpretive conclusions. All AI-generated outputs were critically reviewed, verified against primary sources, and revised by the author. The complete analysis pipeline, including all AI-assisted code, is publicly available in the replication repository (see Data Availability) for full transparency and reproducibility.

Acknowledgments

Author's Premises

This paper was written by a non-specialist who is not a professional researcher. This is not modesty but a structural constraint relevant to interpreting the work.

Advantages of this position: (1) the freedom to propose cross-domain integration that field specialists find difficult; (2) the motivation to construct a testable framework rather than defend a particular paradigm; (3) the lack of career incentives tied to specific conclusions.

Disadvantages: (1) blind spots in domain-specific methodological nuances; (2) possible insufficient grasp of unpublished knowledge; (3) limited peer network for early-stage feedback. These limitations are partially mitigated by the open-source replication repository that allows method-level verification.

As an independent researcher without institutional affiliation or graduate training in the relevant disciplines, I relied on large language models as computational research assistants—much as a solo practitioner might rely on a team of RAs. The specific scope of AI use is documented in the Ethics Statement. I consider this collaboration itself a partial answer to the cognitive asymmetry this paper describes: the same tools that can flood cognitive input can also, under deliberate supervision, amplify an individual's capacity for systematic inquiry. Whether this particular attempt succeeds is for the reader to judge.

This paper is a working paper. I welcome all forms of feedback: methodological corrections, overlooked literature, alternative interpretations, and collaboration proposals.

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Data Availability and Replication Materials

All analysis scripts (15 Python + 1 R), data download procedures, alternative estimators (Driscoll-Kraay SE, Pesaran CCE, Bai IFE, first-difference IV), survey-weighted and covariate-adjusted variants, ad proxy validation data, BibTeX references, publication-ready figures, and paper-to-code correspondence tables are available at:

https://github.com/miyauchikazuyoshi/cognitive-obesity-replication

A Data Specifications

A.1 NHANES 2017–2018 Specifications

Data source: CDC NHANES 2017–2018 cycle

Files used: DEMO_J.XPT, DPQ_J.XPT, PAQ_J.XPT, BMX_J.XPT

Sample: Adults 18+ with complete PHQ-9 responses ($N = 5{,}032$)

Outcome: PHQ-9 sum score (0–27; clinical depression ≥ 10)

Key predictor: Leisure physical exercise (binary: "vigorous/moderate recreational activity" in past 30 days)

Covariates: Age, sex, education (5-level), poverty-income ratio, health insurance status, BMI

Exclusion: Missing PHQ-9, age < 18

A.2 ATUS Wellbeing Module Specifications

Data source: Bureau of Labor Statistics, American Time Use Survey Wellbeing Module (2010, 2012, 2013)

Sample: Adults 15+ with complete Cantril ladder responses ($N = 21{,}736$)

Outcome: Cantril life satisfaction ladder (0–10)

Activity classification: Passive leisure (T120303+T120306), Active cognitive leisure (T120101, T1202xx, T120307–T120313, T1501xx, T150201), Physical exercise (T130101–T130199, T130232)

Covariates: Age, sex, education, household income

A.3 ATUS Activity-to-Loop-Closure Codebook

Table 16 maps ATUS activity codes to the loop-closure categories used in the balance model. Classification is based on whether the activity involves bidirectional agent–environment coupling (closed loop), unidirectional information reception (open loop), or self-referential cognitive processing.

Table 16: ATUS activity code mapping to loop-closure categories.

Category	ATUS codes	Examples	Loop type
Passive leisure	T120303	TV watching	Open
(cognitive input proxy)	T120306	Computer use for leisure	Open
Active cognitive leisure	T120101	Socializing and communicating	Partial
	T1202xx	Attending events (sports, arts)	Partial
(experiential: cognitive)	T120307–T120313	Games, hobbies, reading, music	Partial
	T1501xx	Volunteering	Closed
	T150201	Religious activities	Partial
Physical exercise	T130101–T130199	Sports and exercise	Closed
(experiential: bodily)	T130232	Walking for exercise	Closed

Open = unidirectional input; Partial = bidirectional but cognitively mediated;
Closed = bidirectional with immediate sensorimotor feedback.

Classification rationale. The critical criterion is *whether the activity generates prediction errors from the external environment that update the agent's internal model* (section 2.4). TV/PC leisure involves minimal agent-driven environmental change (open loop). Socializing and games involve bidirectional exchange but are primarily cognitively mediated (partial closure). Sports and exercise involve continuous proprioceptive feedback and motor adjustment (full closure). This three-level coding was applied by a single coder to ATUS tier-2 activity categories; future work should establish inter-rater reliability using independent coders and adjudication protocols.

A.4 Minimum Viable Loop-Closure Annotation Rubric

To facilitate adoption in multimodal systems research and cross-study comparability, we propose a minimum viable rubric for scoring loop closure across any activity or interaction modality (table 17). This rubric is designed to be applicable to time-use surveys, ESM, digital interaction logs, and AI dialogue transcripts.

Table 17: Minimum viable loop-closure scoring rubric.

Dimension	**Open (0)**	**Partial (1)**	**Closed (2)**
Motor engagement	None (seated, passive)	Minimal (typing, tapping)	Continuous (walking, sports)
Environmental feedback	None (broadcast)	Delayed/symbolic (chat reply)	Immediate/physical (ball, tool)
Action contingency	Actions do not alter input	Actions partially alter input	Actions directly shape input
Prediction error source	Internal (rumination)	Mixed (social, game)	External (physical environment
Self-referential content	High (affirmation, echo)	Mixed	Low (task-focused)

Composite score = sum of 5 dimensions (range 0–10). Open: 0–3; Partial: 4–6; Closed: 7–10.
Coding unit: 15-minute epoch (time-use) or single interaction turn (dialogue).

This 5-dimension × 3-level rubric produces a composite score (0–10) that maps onto the three-category system used in the ATUS analysis. The rubric is intentionally minimal: it can be applied by trained coders in under 30 seconds per epoch, and its dimensions are directly operationalizable from passive sensing data (motor engagement via accelerometry), interaction logs

(action contingency via touch events), and self-report (prediction error source). For AI dialogue transcripts, "motor engagement" is replaced by "behavioral activation compliance" (whether the AI prompts real-world action), and "environmental feedback" by "external-referent challenge" (whether the AI introduces information that contradicts the user's model). We encourage researchers to report inter-rater reliability (Cohen's κ or ICC) on the composite score and to share annotated corpora to enable rubric refinement.

A.5 Multiple Testing Correction: FDR Results

Table 18: Multiple Testing Correction: Benjamini–Hochberg FDR

Family	Specification	Raw p	BH q	Bonferroni p
Macro: Depression	AdProxy / Pooled OLS	0.0780	0.1040	0.6243
	AdProxy / Entity FE	$< 10^{-10}$***	$< 10^{-10}$***	$< 10^{-10}$***
	AdProxy / TWFE	0.1811	0.2069	1.0000
	AdProxy / First Difference	1.5×10^{-7}***	4.1×10^{-7}***	1.2×10^{-6}***
	Internet % / Pooled OLS	9.0×10^{-4}***	0.0014**	0.0072**
	Internet % / Entity FE	$< 10^{-10}$***	$< 10^{-10}$***	$< 10^{-10}$***
	Internet % / TWFE	0.3319	0.3319	1.0000
	Internet % / First Difference	4.5×10^{-5}***	9.1×10^{-5}***	3.6×10^{-4}***
Macro: Suicide	AdProxy / Pooled OLS	0.5641	0.5641	1.0000
	AdProxy / Entity FE	1.5×10^{-5}***	1.2×10^{-4}***	1.2×10^{-4}***
	AdProxy / TWFE	0.0070**	0.0112*	0.0559
	AdProxy / First Difference	2.2×10^{-4}***	5.8×10^{-4}***	0.0017**
	Internet % / Pooled OLS	0.1885	0.2154	1.0000
	Internet % / Entity FE	5.1×10^{-5}***	2.1×10^{-4}***	4.1×10^{-4}***
	Internet % / TWFE	0.1558	0.2077	1.0000
	Internet % / First Difference	0.0033**	0.0067**	0.0267*
Macro: DALYs	AdProxy / Pooled OLS	0.0677	0.0902	0.5413
	AdProxy / Entity FE	$< 10^{-10}$***	$< 10^{-10}$***	$< 10^{-10}$***
	AdProxy / TWFE	0.4351	0.4972	1.0000
	AdProxy / First Difference	1.5×10^{-7}***	3.9×10^{-7}***	1.2×10^{-6}***
	Internet % / Pooled OLS	4.0×10^{-4}***	6.4×10^{-4}***	0.0032**
	Internet % / Entity FE	$< 10^{-10}$***	$< 10^{-10}$***	$< 10^{-10}$***
	Internet % / TWFE	0.7322	0.7322	1.0000
	Internet % / First Difference	5.0×10^{-5}***	1.0×10^{-4}***	4.0×10^{-4}***
Placebo	Cardiovascular / TWFE	4.3×10^{-9}***	9.5×10^{-9}***	1.7×10^{-8}***
	Cardiovascular / First Diff	4.7×10^{-9}***	9.5×10^{-9}***	1.9×10^{-8}***
	Diabetes / TWFE	0.9606	0.9606	1.0000
	Diabetes / First Diff	0.7280	0.9606	1.0000
Micro	NHANES: Exercise → PHQ-9	2.2×10^{-10}***	5.0×10^{-10}***	1.5×10^{-9}***
	ATUS: exercise_min → Cantril	6.7×10^{-7}***	1.2×10^{-6}***	4.7×10^{-6}***
	ATUS: active_cog_min → Cantril	0.3573	0.3573	1.0000
	ATUS: passive_min → Cantril	$< 10^{-10}$***	$< 10^{-10}$***	$< 10^{-10}$***
	ATUS: Exercise (binary) → Cantril	1.7×10^{-5}***	2.4×10^{-5}***	1.2×10^{-4}***
	ATUS: Active leisure (binary) → Cantril	0.0067**	0.0078**	0.0470*
	ATUS: High passive (binary) → Cantril	$< 10^{-10}$***	$< 10^{-10}$***	$< 10^{-10}$***

Note. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$. BH q-values computed within each family using the Benjamini–Hochberg step-up procedure. 23 of 35 tests significant before and after BH-FDR correction; 22 survive Bonferroni.